A Bi-objective Priority-Weighted Latency Location-Routing Problem for Pharmaceutical Distribution

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Abstract

Pharmaceutical distribution requires integrated depot-location and vehicle-routing decisions that account for both distribution cost and the urgency of pharmacy replenishment. This study introduces a bi-objective priority-weighted latency location-routing problem (LLRP) that minimizes financial cost and pharmacy-priority-weighted cumulative arrival time under depot, fleet, capacity and route-duration constraints. The objectives are combined through an anchor-normalized augmented weighted Tchebycheff scalarization, which can reach unsupported Pareto-optimal solutions. The problem is solved by a hybrid greedy randomized adaptive search procedure with variable neighborhood descent and strategic oscillation (GRASP-VND-SO), whose solutions are validated against exact methods based on a priority-flow reformulation and branch-and-price. On ten standard LLRP benchmarks, GRASP-VND-SO attains the best-known solution on six instances, with a mean gap of 0.46%. On four pharmaceutical networks in Türkiye under five priority profiles, it attains the certified optimum at all 45 interior preferences of the smallest network but is less reliable on the most dispersed one. At the balanced preference, the priority-weighted latency is 12.9–52.8% lower than at the cost anchor, with travel cost rising by at most 18.4%. Concentrating weights on a few critical pharmacies moves them to the beginning of their routes, whereas weights spread over many pharmacies change the service only slightly. The open depots change with both the preference and the priority profile.

1. Introduction

Pharmaceutical distribution combines financial logistics decisions with the need for timely pharmacy replenishment. The latency location-routing problem (LLRP) jointly determines facility locations and vehicle routes while minimizing cumulative customer arrival times. However, treating every customer's waiting time as equally important may not adequately represent differentiated pharmacy-service priorities.

This study develops a bi-objective priority-weighted LLRP that minimizes financial distribution cost and priority-weighted cumulative arrival time. Depot opening, pharmacy assignment, vehicle deployment, and delivery sequence decisions are optimized jointly subject to capacity, fleet, and route-duration constraints. An anchor-normalized augmented weighted Tchebycheff formulation explores cost–service compromises without requiring an arbitrary monetary conversion of waiting time and, unlike a weighted sum, can also reach unsupported Pareto-optimal solutions.

To address the problem's combinatorial complexity, we develop a hybrid greedy randomized adaptive search procedure (GRASP) with variable neighborhood descent (VND) and strategic oscillation (SO), hereafter GRASP-VND-SO. Construction based on the Extended Clarke–Wright algorithm (ECWA), seven route-level neighborhoods, and depot replacement are integrated within one search framework. The operators build on established LLRP search mechanisms; their contribution is examined in a component study on standard benchmarks. To validate the heuristic, the problem is also solved exactly with a commodity-flow reformulation, in which a flow of priority weights represents the priority-weighted latency without big-M constraints, and with a branch-and-price algorithm for the harder anchor problems.

The contributions are organized around three complementary empirical evaluations. First, the established LLRP search architecture is adapted to the priority-weighted bi-objective setting and assessed against published algorithms on standard unweighted-latency benchmarks, supported by a component analysis of the search configurations. Second, four real pharmaceutical distribution networks are used to quantify how alternative pharmacy priorities change the cost–arrival-time compromises and the location decisions within the same location-routing constraints. Third, the heuristic solutions on these networks are validated against exact solutions: exact anchor values are computed for three of the four networks, and all interior preferences of the smallest network are solved to proven optimality. The methodological contribution is the integrated priority-sensitive formulation, its priority-flow reformulation, and its reproducible multi-objective computational application, rather than a claim that the individual GRASP, VND, or SO operators are new.

The remainder is organized as follows. Section 2 reviews related research; Section 3 specifies the problem, formulation, and scalarization; Section 4 presents GRASP-VND-SO; Section 5 addresses benchmarking, validation against exact solutions and the pharmaceutical experiments; and Section 6 concludes. Proofs, the branch-and-price algorithm, details of the heuristic and additional results are given in the appendices.

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2. Literature Review

The location-routing problem (LRP) integrates facility selection and vehicle routing, avoiding inefficiencies inherent in sequential decisions (Salhi and Rand, 1989). Foundational surveys by Min et al. (1998); Nagy and Salhi (2007) and subsequent taxonomies (Lopes et al., 2013; Prodhon and Prins, 2014; Drexl and Schneider, 2015; Schneider and Drexl, 2017; Mara et al., 2021; Cavagnini et al., 2026a) document its structural variants and solution methods. Deterministic LRPs commonly assume single-period demand and single-visit customers. Their combinatorial complexity has motivated simulated annealing, particle swarm optimization, tree-based search, progressive filtering, and hybrid randomized local-search approaches (Asefi et al., 2017; Ahmed and Sun, 2018; Schneider and Löffler, 2019; Arnold and Sörensen, 2021; Löffler et al., 2023).

The LLRP replaces conventional distance-oriented service evaluation with cumulative customer arrival time. Moshref-Javadi and Lee (2016) introduced mathematical and metaheuristic approaches, followed by branch-and-cut and GRASP-based iterated local search (Nucamendi-Guillén et al., 2022), simulated annealing with VND (Osorio-Mora et al., 2023), reinforcement-learning-guided evolutionary search (Zou et al., 2024), risk-averse travel-time modeling (Osorio-Mora et al., 2025), and split-delivery extensions (Nucamendi-Guillén et al., 2026). Multi-objective location-routing models are reviewed by Cavagnini et al. (2026b). To our knowledge, no location-routing study combines pharmacy-specific priority-weighted cumulative arrival time with financial cost in a bi-objective formulation.

Service differentiation is established in priority-sensitive routing (Doan et al., 2021), healthcare applications (Akbari et al., 2023), and emergency location-routing (Oran et al., 2012). Humanitarian routing distinguishes efficiency, efficacy, and equity (Huang et al., 2012); deprivation-cost models represent the consequences of delayed deliveries (Holguín-Veras et al., 2013), while inequity measures and explicit emergency priorities address different policy objectives (Gutjahr and Fischer, 2018; Zhu et al., 2019). Unlike these concepts, priority-weighted latency preserves cumulative arrival time while varying the importance assigned to each customer's delay. It neither imposes service deadlines nor directly measures equity.

Location-routing models have also been applied to pharmaceutical and healthcare distribution. Ceselli et al. (2014) coordinate depots, distribution centers and vehicles for the distribution of vaccines or drugs and solve the problem exactly by branch-and-price, and Veenstra et al. (2018) decide which medication pick-up lockers to open while routing vehicles to the lockers and to the remaining patients. Cedolin and Bal (2026) allocate pharmaceutical warehouses under stochastic demand with priority given to critical medicines. These studies do not differentiate the urgency of individual delivery points through priority-weighted arrival times.

Hybrid metaheuristics provide a relevant computational foundation. GRASP and related hybrids have been applied to capacitated location-routing and location-arc routing problems (Hamidi et al., 2014; Ferdi and Layeb, 2018; Löffler et al., 2023; Lopes et al., 2014); VND and multi-objective neighborhood methods have also been used in location-routing (Jabal-Ameli et al., 2011; Ghaffari-Nasab et al., 2013; Leite et al., 2022). Within LLRP research, Osorio-Mora et al. (2023) and Zou et al. (2024) further demonstrate the value of complementary neighborhoods.

The present contribution lies in combining financial cost, pharmacy-priority-weighted cumulative latency, and integrated depot and vehicle-routing decisions, using normalized multi-objective optimization, a common GRASP-VND-SO architecture, and exact validation based on a priority-flow reformulation and branch-and-price.

3. Problem Definition and Mathematical Formulation

3.1. Problem Description

This study addresses a priority-weighted LLRP that integrates strategic depot-location decisions and operational vehicle-routing decisions. Given a set of candidate depot locations, the problem determines which N_g depots are opened, how pharmacies are assigned to the selected depots, and how vehicle routes are constructed to serve all pharmacies, taking into account both financial distribution cost and priority-sensitive delivery performance.

The problem is defined on a directed graph $G = (V, A)$, where $V = I \cup J$ is the complete set of nodes. The set I represents candidate depot locations, whereas J represents pharmacies. The arc set $A = \{(u, v) : u, v \in V, u \neq v\} \setminus (I \times I)$ excludes direct depot-to-depot movements. Each arc $(u, v) \in A$ is associated with a travel cost c_{uv}, derived from road distance, and a travel duration ℓ_{uv}; both matrices are directed, and neither symmetry nor the triangle inequality is imposed as a modeling assumption.

Each pharmacy $j \in J$ has a deterministic demand d_j and must be visited exactly once by a route originating from an open depot. At most N_v homogeneous vehicles of capacity Q are available across the network, and each candidate depot $i \in I$ can serve a total assigned demand of at most W_i. Every route starts from and returns to the same open depot, and every open depot dispatches at least one vehicle. The total travel duration of each route, including the return trip to the depot, cannot exceed the maximum route duration L.

All routes start at time zero when the vehicle leaves its depot, and no service times are considered. The arrival time a_j of pharmacy j is therefore the cumulative travel duration from the depot to j along its route. Each pharmacy is assigned a positive priority coefficient p_j, where larger values indicate that its delay is more important, and service performance is measured by the priority-weighted latency $\sum_{j \in J} p_j a_j$. Equal units of waiting time can therefore have different consequences at different pharmacies. Because latency is incurred upon arrival at the pharmacies, the return trip to the depot affects route-duration feasibility but not latency.

The problem has two conflicting criteria, evaluated over the same set of feasible location-routing decisions: the total financial cost, comprising depot-opening, vehicle-deployment and transportation costs, and the total priority-weighted latency. The two criteria are combined through the scalarization described in Section 3.5.

3.2. Model Assumptions

The model is developed for a single planning period, corresponding to one distribution shift. Pharmacy demands, priority coefficients, travel costs and travel durations are known in advance and remain unchanged within this period, so congestion and other time-dependent effects are not represented. The demand of each pharmacy is delivered in full in a single visit, split deliveries are not permitted, and each vehicle performs at most one route during the period.

Depots can be opened only at predefined candidate locations, and the number of open depots is fixed, while their locations are decision variables. The fleet of N_v vehicles is shared by the network, and each vehicle is assigned to the open depot from which it operates. Opening costs, vehicle-deployment costs and depot capacities may differ across depots in the model, although they are identical across candidate depots in the case studies of Section 5. All vehicles have the same capacity Q and are subject to the same route-duration limit L.

The priority coefficients are exogenous and fixed before optimization. Each priority profile considered in Section 5 represents a different service policy, and the coefficients do not change within a scenario. Delivery urgency is represented only through the priority-weighted latency; time windows, delivery deadlines and penalties for late service are not imposed.

The mathematical notation is summarized in Appendix A (Table A.1).

3.3. Mathematical Formulation

Mathematical formulations of the LRP differ mainly in how routing flows are represented and how subtours are eliminated (Prodhon and Prins, 2014; Schneider and Drexl, 2017). Formulations with exponentially many subtour-elimination constraints or with multi-commodity flows provide strong linear relaxations, but they require separation procedures or grow rapidly with the size of the network. We use a compact depot-indexed (three-index) vehicle-flow formulation, in which the routing variable carries the index of the depot that operates the route, so that each route remains attached to a single depot without route-level variables (Moshref-Javadi and Lee, 2016; Nucamendi-Guillén et al., 2022). Subtours are eliminated by Miller–Tucker–Zemlin (MTZ) type constraints (Miller et al., 1960), applied to both the cumulative load and the arrival time. Since MTZ constraints are known to yield weak linear relaxations, Section 3.4 discusses how the formulation is strengthened.

Let $A_J = \{(u,v) : u,v \in J, u \neq v\}$ denote the set of arcs between pharmacies, and for each depot $i \in I$ let $A^i = \{(i,j),(j,i) : j \in J\} \cup A_J$ denote the arcs that a route operated from depot i may use. The binary variable y_i equals 1 if depot i is opened, z_{ij} equals 1 if pharmacy j is assigned to depot i, and x^i_{uv} equals 1 if arc $(u,v) \in A^i$ is traversed by a route of depot i. The integer variable k_i is the number of vehicles dispatched from depot i. The continuous variables ω_j and t_j are the cumulative load and the arrival time at pharmacy j; the load variables enforce the vehicle capacity, and the arrival-time variables enforce the route-duration limit and define the latency criterion.

The first objective is the total financial cost, consisting of depot-opening, transportation and vehicle-deployment costs:

$$C_F = \sum_{i\in I} f_i y_i + \sum_{i\in I}\sum_{(u,v)\in A^i} c_{uv} x^i_{uv} + \sum_{i\in I} F_i k_i. \tag{1}$$

The second objective is the total priority-weighted latency:

$$P = \sum_{j\in J} p_j t_j. \tag{2}$$

The two objectives are measured in different units and are therefore not added directly. Instead, they are normalized with respect to anchor values and combined through an augmented weighted Tchebycheff scalarization, described in Section 3.5, which allows the trade-off between cost and priority-weighted latency to be explored without assigning a monetary value to delay. The bi-objective model is stated below. Let S denote the set of feasible solutions, that is, the depot, assignment and routing decisions satisfying constraints (4)–(25). For a solution $S \in \mathcal{S}$, $C_F(S)$ denotes its financial cost (1), and $P(S) = \sum_{j\in J} p_j a_j(S)$ denotes its priority-weighted latency, where $a_j(S)$ is the actual arrival time of pharmacy j on its route; $P(S)$ coincides with (2) when the arrival-time variables are tight.

$$\min \quad \big(C_F,\ P\big) \tag{3}$$

s.t.

$$\sum_{i\in I} z_{ij} = 1 \qquad \forall j \in J \tag{4}$$

$$\sum_{j\in J} d_j z_{ij} \le W_i y_i \qquad \forall i \in I \tag{5}$$

$$z_{ij} \le y_i \qquad \forall i \in I,\ j \in J \tag{6}$$

$$\sum_{i\in I} y_i = N_g \tag{7}$$

$$\sum_{\substack{u\in J\cup\{i\}\\(u,j)\in A^i}} x^i_{uj} = z_{ij} \qquad \forall i \in I,\ j \in J \tag{8}$$

$$\sum_{\substack{v\in J\cup\{i\}\\(j,v)\in A^i}} x^i_{jv} = z_{ij} \qquad \forall i \in I,\ j \in J \tag{9}$$

$$\sum_{j\in J} x^i_{ij} = k_i \qquad \forall i \in I \tag{10}$$

$$\sum_{j\in J} x^i_{ji} = k_i \qquad \forall i \in I \tag{11}$$

$$y_i \le k_i \le N_v y_i \qquad \forall i \in I \tag{12}$$

$$\sum_{i\in I} k_i \le N_v \tag{13}$$

$$\sum_{j\in J} d_j z_{ij} \le Q k_i \qquad \forall i \in I \tag{14}$$

$$\omega_v - \omega_u \geq d_v - Q\Big(1 - \sum_{i\in I} x^i_{uv}\Big) \qquad \forall (u,v) \in A_J \qquad (15)$$

$$d_j \leq \omega_j \leq Q \qquad \forall j \in J \qquad (16)$$

$$t_v - t_u \geq \ell_{uv} - M^T_{uv}\Big(1 - \sum_{i\in I} x^i_{uv}\Big) \qquad \forall (u,v) \in A_J \qquad (17)$$

$$t_j \geq \sum_{i\in I} \ell_{ij} x^i_{ij} \qquad \forall j \in J \qquad (18)$$

$$0 \leq t_j \leq L \qquad \forall j \in J \qquad (19)$$

$$t_j + \sum_{i\in I} \ell_{ji} x^i_{ji} \leq L \qquad \forall j \in J \qquad (20)$$

$$y_i \in \{0,1\} \qquad \forall i \in I \qquad (21)$$

$$z_{ij} \in \{0,1\} \qquad \forall i \in I,\ j \in J \qquad (22)$$

$$x^i_{uv} \in \{0,1\} \qquad \forall i \in I,\ (u,v) \in A^i \qquad (23)$$

$$k_i \in \mathbb{Z}_{\geq 0} \qquad \forall i \in I \qquad (24)$$

$$\omega_j \geq 0, \quad t_j \geq 0 \qquad \forall j \in J. \qquad (25)$$

Constraints (4) assign each pharmacy to exactly one depot, and constraints (5) limit the demand assigned to depot i to its capacity W_i. Constraints (6) allow assignment only to open depots; since all demands are strictly positive, they are implied by (5) for integer solutions, but they are retained because they strengthen the linear relaxation. Constraint (7) fixes the number of open depots at N_g.

Constraints (8) and (9) require each pharmacy assigned to depot i to have exactly one incoming and one outgoing arc in the routing layer of that depot. Constraints (10) and (11) set k_i equal to both the number of departures from and the number of returns to depot i, so that every route begins and ends at the same depot. Constraints (12) prevent closed depots from operating vehicles and require every open depot to dispatch at least one vehicle, and constraint (13) limits the total fleet to N_v vehicles. Constraints (14) are aggregate capacity inequalities linking the demand assigned to a depot with the number of vehicles it dispatches; route-level capacity is enforced by the load-propagation constraints.

Constraints (15) and (16) propagate the cumulative load along pharmacy-to-pharmacy arcs and bound it by the vehicle capacity, which enforces vehicle capacity and eliminates subtours among pharmacies. Constraints (17)–(20) play the same role for arrival times. Constraints (17) propagate arrival times along pharmacy-to-pharmacy arcs; with $M^T_{uv} = L + \ell_{uv}$ they become redundant when arc (u,v) is not used, because $t_u \leq L$ and $t_v \geq 0$. Constraints (18) initialize the arrival time of the first pharmacy on a route, and constraints (19) bound all arrival times by L. Constraints (20) impose the duration limit on the return trip: if pharmacy j is the last pharmacy of a route returning to depot i, then $t_j + \ell_{ji} \leq L$, and otherwise the constraint reduces to $t_j \leq L$. The return trip therefore counts toward route duration without contributing to latency.

Because at most one routing layer can use a given arc, the propagation constraints are written for the aggregated flow $\sum_{i\in I} x^i_{uv}$ rather than separately for each depot. This reduces their number by a factor of $|I|$, and the aggregated form is at least as tight in the linear relaxation, since $\sum_{i\in I} x^i_{uv} \geq x^i_{uv}$ for every i. Since each pharmacy is served from exactly one depot, ω_j and t_j are defined at the pharmacy level rather than for each depot–pharmacy pair, and depot consistency is maintained by the depot-indexed variables x^i_{uv} together with the assignment and degree constraints. Constraints (21)–(25) define the variable domains.

Since the arrival-time constraints bound t_j only from below, $t_j \geq a_j(S)$ for every feasible solution, with equality at an optimum whenever P is minimized with a positive weight. When only C_F is minimized, the arrival-time variables need not be tight; the latencies reported in this paper are therefore always computed from the routes as $P(S)$.

3.4. Alternative Formulation and Strengthening

The MTZ constraints of formulation (3)–(25) yield a weak linear relaxation, and in the present problem the weakness affects the objective directly. The arrival-time variables are bounded from below only through the propagation constraints (17)–(18), whose big-M terms are easily satisfied by fractional routing solutions, so the latency term of the relaxation can be far below its value in any integer solution. This subsection presents a commodity-flow reformulation that represents both the vehicle load and the priority-weighted latency without big-M constraints, followed by arrival-time windows, preprocessing rules, and a valid inequality. None of these modifications changes the set of feasible integer solutions or their objective values.

For each arc $(u,v) \in A_J$, let $X_{uv} = \sum_{i\in I} x^i_{uv}$ denote the aggregated arc flow. The load is represented by a single-commodity flow (Gavish and Graves, 1978): $g_{ij} \geq 0$ is the load carried on depot arc (i,j) and $g_{uv} \geq 0$ the load carried on arc $(u,v) \in A_J$. The constraints

$$\sum_{i\in I} g_{ij} + \sum_{(u,j)\in A_J} g_{uj} - \sum_{(j,v)\in A_J} g_{jv} = d_j \qquad \forall j \in J \qquad (26)$$

$$d_j\, x^i_{ij} \leq g_{ij} \leq Q\, x^i_{ij} \qquad \forall i \in I,\ j \in J \qquad (27)$$

$$d_v\, X_{uv} \leq g_{uv} \leq Q\, X_{uv} \qquad \forall (u,v) \in A_J \qquad (28)$$

replace the load-propagation constraints (15)–(16). Each pharmacy absorbs its own demand, so the flow leaving a depot equals the total demand of the route and is limited by Q. A cycle of pharmacies that does not contain a depot cannot satisfy (26), because summing the constraints around the cycle gives zero on the left-hand side and a positive total demand on the right-hand side; the flow therefore also eliminates subtours.

The latency is represented in the same way by a flow of priority. Let $r_{ij} \geq 0$ and $r_{uv} \geq 0$ denote the priority carried on depot arc (i,j) and on arc $(u,v) \in A_J$, interpreted as the total priority of the pharmacies that are still to be served when the vehicle traverses the arc, and let $\Pi = \sum_{j\in J} p_j$. The constraints

$$\sum_{i\in I} r_{ij} + \sum_{(u,j)\in A_J} r_{uj} - \sum_{(j,v)\in A_J} r_{jv} = p_j \qquad \forall j \in J \qquad (29)$$

$$p_j\, x^i_{ij} \le r_{ij} \le \Pi\, x^i_{ij} \qquad \forall i \in I,\ j \in J \tag{30}$$

$$p_v\, X_{uv} \le r_{uv} \le \Pi\, X_{uv} \qquad \forall (u,v) \in A_J \tag{31}$$

define the priority-flow latency

$$P^F = \sum_{i\in I}\sum_{j\in J} \ell_{ij}\, r_{ij} + \sum_{(u,v)\in A_J} \ell_{uv}\, r_{uv}. \tag{32}$$

No priority flow is defined on arcs returning to a depot, consistent with latency being incurred only upon arrival at the pharmacies.

Proposition 1. *For every feasible integer routing solution S, constraints (29)–(31) have a unique solution, and $P^F(S) = \sum_{j\in J} p_j a_j(S) = P(S)$.*

Proof. Consider a route $(i, j_1, \dots, j_h, i)$ and set $j_0 = i$. By (30)–(31), the priority flow is zero on every unused arc, and each pharmacy j_s has exactly one incoming and one outgoing used arc. The outgoing arc of j_h returns to the depot and carries no priority flow, so (29) at j_h gives $r_{j_{h-1}j_h} = p_{j_h}$. Proceeding backward along the route, the flow on the arc entering j_s is $r_{j_{s-1}j_s} = \sum_{q=s}^{h} p_{j_q}$ for $s = 1, \dots, h$, which also satisfies the bounds in (30)–(31). Hence the contribution of the route to (32) is

$$\sum_{s=1}^{h} \ell_{j_{s-1}j_s} \sum_{q=s}^{h} p_{j_q} = \sum_{q=1}^{h} p_{j_q} \sum_{s=1}^{q} \ell_{j_{s-1}j_s} = \sum_{q=1}^{h} p_{j_q} a_{j_q}(S).$$

Summing over all routes gives the result.

Unlike $\sum_{j\in J} p_j t_j$, the priority-flow latency equals the actual latency of every integer solution, independently of the tightness of the arrival-time variables, and it links the latency to the movement of priority weights along the selected arcs instead of big-M constraints. Flow-based representations of cumulative objectives have been used for the traveling deliveryman problem and the cumulative capacitated vehicle routing problem (Méndez-Díaz et al., 2008; Lysgaard and Wøhlk, 2014); here the flow carries pharmacy-specific priorities. The arrival-time variables and constraints (17)–(20) are retained to enforce the route-duration limit, while (32) evaluates the latency.

The arrival-time constraints can be tightened further. Let $\underline{\ell}_{uv}$ denote the duration of a shortest directed path from node u to node v. Shortest-path durations are required because the road-network durations need not satisfy the triangle inequality, so a direct arc is not necessarily the fastest connection. Since any route reaches pharmacy j no earlier than $e_j = \min_{i\in I} \underline{\ell}_{ij}$ and must still return to a depot afterward, the bounds

$$e_j \le t_j \le L - \min_{i\in I} \underline{\ell}_{ji} =: L_j \qquad \forall j \in J \tag{33}$$

are valid, and the big-M coefficient in (17) can be reduced to $M^T_{uv} = \max\{0,\ \ell_{uv} + L_u - e_v\}$.

The formulation can also be reduced in size before optimization. The assignment of pharmacy j to depot i can be eliminated if

$$d_j > Q, \qquad d_j > W_i, \qquad \text{or} \qquad \underline{\ell}_{ij} + \underline{\ell}_{ji} > L, \tag{34}$$

since the pharmacy then cannot be served by a vehicle, cannot be accommodated by the depot, or cannot be reached and left within the duration limit even along shortest paths. Similarly, the transition $u \to v$ can be removed from the routes of depot i if

$$d_u + d_v > \min\{Q, W_i\} \qquad \text{or} \qquad \underline{\ell}_{iu} + \ell_{uv} + \underline{\ell}_{vi} > L, \tag{35}$$

because any route containing both pharmacies carries at least their combined demand, and the shortest-path durations give a lower bound on the duration of any route that contains the transition. Finally, since every route carries at most Q units of demand, the fleet inequality

$$\sum_{i\in I} k_i \ge \left\lceil \frac{\sum_{j\in J} d_j}{Q} \right\rceil \tag{36}$$

is valid and strengthens the relaxation when vehicle capacity is tight.

3.5. Augmented Weighted Tchebycheff Scalarization

The financial cost and the priority-weighted latency are measured in different units, and no reliable monetary value of a pharmacy's waiting time is available, so the two criteria are not added directly. A weighted sum of normalized objectives would avoid the unit problem, but its optima are restricted to supported efficient solutions, which lie on the boundary of the convex hull of the objective space; efficient solutions in nonconvex parts of the Pareto front cannot be obtained for any choice of weights (Das and Dennis, 1997; Ehrgott, 2005). Such unsupported solutions are common in discrete problems such as location routing. The weighted Tchebycheff scalarization does not have this limitation: every Pareto-optimal solution is optimal for some choice of weights, and the augmented form, with a sufficiently small augmentation coefficient, excludes weakly efficient solutions (Steuer and Choo, 1983; Miettinen, 1999). The trade-off parameter of the normalized function is dimensionless and expresses a relative preference between the two criteria rather than a price of delay.

Normalization is based on the two lexicographic anchor solutions of the bi-objective problem. For a given instance and priority profile, the anchor values are

$$\widehat{C}_{\min} = \min_{S\in\mathcal{S}} C_F(S), \tag{37}$$

$$\widehat{P}_C = \min_{S\in\mathcal{S}} \big\{ P(S) \,:\, C_F(S) = \widehat{C}_{\min} \big\},$$

$$\widehat{P}_{\min} = \min_{S\in\mathcal{S}} P(S), \tag{38}$$

$$\widehat{C}_P = \min_{S\in\mathcal{S}} \big\{ C_F(S) \,:\, P(S) = \widehat{P}_{\min} \big\},$$

so that $(\widehat{C}_{\min}, \widehat{P}_{\min})$ is the ideal point and, for two objectives, $(\widehat{C}_P, \widehat{P}_C)$ is the nadir point of the Pareto front (Ehrgott, 2005). With the ranges $\Delta_C = \widehat{C}_P - \widehat{C}_{\min} > 0$ and $\Delta_P = \widehat{P}_C - \widehat{P}_{\min} > 0$, the normalized criteria

$$\overline{C}(S) = \frac{C_F(S) - \widehat{C}_{\min}}{\Delta_C}, \qquad \overline{P}(S) = \frac{P(S) - \widehat{P}_{\min}}{\Delta_P} \tag{39}$$

take values in $[0, 1]$ for every Pareto-optimal solution.

When an anchor is obtained heuristically, its first-stage value is retained for normalization. Solutions found subsequently may improve upon this reference value, resulting in slightly negative normalized values without affecting the monotonicity and Pareto-efficiency properties of the scalarization.

When an anchor cannot be certified by an exact method, the best known feasible value over all computations is used; normalized values of later solutions may then be slightly negative, which does not affect the properties below.

For a trade-off parameter $\lambda \in (0, 1)$ and an augmentation coefficient $\rho > 0$, the scalarized objective is

$$\begin{aligned} Z_\lambda(S) = \; & \max\big\{(1-\lambda)\overline{C}(S),\ \lambda\overline{P}(S)\big\} \\ & + \rho\big[(1-\lambda)\overline{C}(S) + \lambda\overline{P}(S)\big]. \end{aligned} \tag{40}$$

The first term measures the larger weighted deviation from the ideal point, and the second term breaks ties in favor of solutions that are better in both criteria. Larger values of λ give more weight to latency. In the computations, $\rho = 10^{-4}$ and $\lambda \in \{0.1, 0.2, \dots, 0.9\}$, or $\lambda \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ for the largest network; the endpoints $\lambda = 0$ and $\lambda = 1$ correspond to the cost and latency anchors.

Proposition 2. *For $0 < \lambda < 1$, $\rho > 0$, $\Delta_C > 0$ and $\Delta_P > 0$, every optimal solution of $\min_{S\in\mathcal{S}} Z_\lambda(S)$ is Pareto efficient with respect to C_F and P.*

The result follows from the strict monotonicity of Z_λ in both criteria (Steuer and Choo, 1983; Ehrgott, 2005); a proof is given in Appendix C. Conversely, since $\mathcal{S}$ is finite, every Pareto-optimal solution is optimal for some λ when ρ is sufficiently small (Steuer and Choo, 1983). The nine values of λ used here therefore yield a representative subset of the Pareto front rather than the complete front, and different values of λ may lead to the same solution. Solutions obtained heuristically are not guaranteed to be Pareto efficient and are subjected to nondominance screening in Section 5.

For fixed anchor values, the scalarized problem is a mixed-integer linear program. With an auxiliary variable $\zeta \in \mathbb{R}$, it reads

$$\begin{aligned} \min \quad & \zeta + \rho\big[(1-\lambda)\overline{C} + \lambda\overline{P}\big] \\ \text{s.t.} \quad & \zeta \geq (1-\lambda)\overline{C}, \quad \zeta \geq \lambda\overline{P}, \quad S \in \mathcal{S}, \end{aligned} \tag{41}$$

where $\overline{C}$ and $\overline{P}$ are affine in the decision variables. If the latency is expressed by the priority flow P^F of Eq. (32), the formulation evaluates the exact latency of every integer solution, independently of the tightness of the arrival-time variables.

Finally, if all priority coefficients are multiplied by a common factor $\gamma > 0$, the lexicographic anchors scale accordingly, the normalized criteria are unchanged, and Z_λ assigns the same value to every solution; the derivation is given in Appendix C.

4. GRASP-VND-SO Heuristic

The heuristic combines GRASP (Feo and Resende, 1995) with VND that uses SO. Each GRASP iteration selects a set of depots, builds routes with a randomized savings procedure, and improves the resulting solution with VND-SO. The improvement phase uses seven route-level neighborhoods and a depot-replacement operator adopted from Osorio-Mora et al. (2023) and Zou et al. (2024); the contribution of this study lies in their integration within GRASP, the capacity-based strategic oscillation, and the adaptation of the search to the priority-weighted and scalarized objectives. SO allows temporary violations of vehicle capacity, but only fully feasible solutions can become incumbents. The heuristic operates in three modes: minimization of financial cost and minimization of priority-weighted latency, which yield the anchor solutions, and minimization of the augmented weighted Tchebycheff function (40) for the interior preferences. The anchor modes minimize a single criterion and do not break ties by the other one. The heuristic anchor values are therefore approximations of the lexicographic values defined in Section 3.5, and the cross values $\widehat{P}_C$ and $\widehat{C}_P$ are taken as the best values recorded at the corresponding anchor cost or latency over all runs.

4.1. Solution Representation and Evaluation

A solution S consists of exactly N_g open depots and at most N_v nonempty routes; each route starts and ends at the same open depot, every open depot operates at least one route, and every pharmacy is visited exactly once. For a route $r = (i, j_1, \dots, j_h, i)$, the arrival times are

$$\begin{aligned} & a_{j_1}(S) = \ell_{ij_1}, \\ & a_{j_s}(S) = a_{j_{s-1}}(S) + \ell_{j_{s-1}j_s}, \qquad s = 2, \dots, h, \end{aligned} \tag{42}$$

and the route duration is $a_{j_h}(S) + \ell_{j_h i}$. The return trip therefore counts toward the duration limit but not toward latency. A solution is feasible if it satisfies the vehicle capacity Q, the depot capacities W_i, the duration limit L, and the fleet limit N_v.

The financial cost $C_F(S)$, the priority-weighted latency $P(S) = \sum_{j\in J} p_j a_j(S)$, and the scalarized objective $Z_\lambda(S)$ are evaluated as defined in Sections 3.3 and 3.5. Both criteria are sums of route contributions, so the effect of a move is computed from the routes it changes: a move that alters the sequence of a route requires recomputing the arrival times downstream of the change and yields the increments ΔC_F and ΔP. The value of the scalarized objective after the move is then obtained directly as Z_λ evaluated at $C_F(S)+\Delta C_F$ and $P(S) + \Delta P$; the maximum in Z_λ is applied to the solution totals, not to individual routes. During strategic oscillation only the vehicle capacity may be violated, and only feasible solutions can become incumbents.

4.2. Constructive Phase: ECWA

Initial solutions are built by an extended version of the savings algorithm of Clarke and Wright (1964) (ECWA).

The construction starts from a set of N_g open depots. Pharmacies are first assigned to depots by a greedy procedure that processes pharmacies with fewer feasible depots first and accounts for vehicle and depot capacities, round-trip duration, and depot utilization; if a pharmacy cannot be assigned feasibly, the construction fails and the iteration produces no solution. Each assigned pharmacy initially forms its own route (i, j, i).

Pairs of routes are then merged, possibly across depots. For two routes r_1 and r_2, both concatenation orders and all four orientation combinations are considered, since the travel matrices are asymmetric, and the merged route r_m is assigned to the depot of r_1 or of r_2, whichever gives the larger saving and respects that depot's capacity. A merge is admissible only if r_m satisfies the vehicle capacity and the duration limit. The saving of a merge is

$$s(r_1, r_2) = \tau(r_1) + \tau(r_2) - \tau(r_m), \tag{43}$$

where $\tau(r)$ is the duration of route r including the return to its depot. The admissible merges $\mathcal{M}_{\text{adm}}$ are ranked by saving, and one merge is drawn uniformly from a restricted candidate list (RCL) containing the

$$|\text{RCL}| = \min\{\kappa_{\text{RCL}}, |\mathcal{M}_{\text{adm}}|\} \tag{44}$$

best merges. Merges with nonpositive savings are allowed only when further merging is needed to respect the fleet limit N_v. The construction ends when no admissible merge remains. The priority coefficients and λ are not used during construction; they enter only through the improvement phase and the selection of incumbents. Further details are given in Appendix E.

4.3. Improvement Phase: VND-SO

Each constructed solution is improved by a variable neighborhood descent with strategic oscillation (Glover and Hao, 2011). The descent uses a depot-replacement operator followed by seven route-level neighborhoods, in the order relocate, swap, 2-opt/2-opt*, 2-relocate, node–arc swap, arc–arc swap, and Swap*; formal definitions are given in Appendix F. The swap neighborhood also exchanges the depots of two routes. All moves preserve customer coverage, the number of open depots, and the number of routes, so the route count of a solution is determined during construction.

4.3.1. Neighborhood Search Strategy

Each route-level neighborhood is searched with a first-improvement strategy: the first admissible improving move is applied, the same neighborhood is searched again, and the next neighborhood is reached only when no improving move remains. Depot replacement, applied at the start of each pass, selects the best improving move instead. For an open depot $i \in D(S)$ and a closed candidate $i' \in I \setminus D(S)$, it keeps the pharmacy sequences of all routes of i and assigns them to i',

$$D(S') = (D(S) \setminus \{i\}) \cup \{i'\}, \tag{45}$$

after which costs, arrival times, depot loads, and route durations are reevaluated. A new pass starts whenever any operator has applied a move, and the descent ends after a pass without improvement.

Strategic oscillation allows the vehicle capacity to be violated temporarily, while depot capacities and the duration limit remain hard constraints. With route load q_r, the capacity violation and the penalized objective are

$$V_Q(S) = \sum_{r \in \mathcal{R}(S)} \max\{0, q_r - Q\}, \tag{46}$$

$$\Phi_{o,\beta}(S) = F_o(S) + \beta V_Q(S), \tag{47}$$

where F_o is the objective of the current mode, defined in (48), and $\beta > 0$ is an adaptive penalty. A move is accepted if it decreases $\Phi_{o,\beta}$ by more than a small tolerance $\varepsilon_{\text{move}}(o)$. After four consecutive accepted moves leading to capacity-infeasible solutions, β is multiplied by a factor drawn at random from $\{1.5, 2.5\}$; after four consecutive feasible ones it is divided by such a factor. The initial value of β is scaled to the objective of the starting solution; the values of all parameters are given in Appendix G. Throughout the descent, the best feasible solution found is kept as the incumbent and is compared using the unpenalized objective F_o; it is returned when the descent ends, even if the final search state is infeasible.

4.3.2. GRASP Framework

The GRASP alternates diversification and intensification. In a diversification iteration, the N_g depots used by ECWA are drawn uniformly at random; after five such iterations, the open depots of the best feasible solution found in them, after improvement, are used for the next three intensification iterations, during which depot replacement can still change them. The cycle then restarts, and an iteration that yields no feasible solution leaves the incumbent unchanged. The objective used to compare feasible solutions in mode $o \in \{\texttt{financial}, \texttt{latency}, \texttt{tchebycheff}\}$ is

$$F_o(S) = \begin{cases} C_F(S), & o = \texttt{financial}, \\ P(S), & o = \texttt{latency}, \\ Z_\lambda(S), & o = \texttt{tchebycheff}, \end{cases} \tag{48}$$

and the mode is fixed throughout a run. The complete procedure is summarized in Algorithm 1, and the VND-SO pseudocode is given in Algorithm 2 (Appendix H).

5. Computational Study

The computational study is organized as follows. Section 5.2 assesses the quality of GRASP-VND-SO on standard LLRP benchmark instances and the contribution of its main search components. After the pharmaceutical instances are introduced in Section 5.3, Section 5.4 measures the distance between heuristic and exact solutions on these networks. Section 5.5 analyzes the trade-off between financial

Algorithm 1 GRASP-VND-SO

Input: instance data, mode o, iterations T, RCL size κ_{RCL}, period lengths T_D, T_I
Output: best feasible solution S_{best}

$S_{\text{best}} \leftarrow \emptyset$; $S_{\text{div}} \leftarrow \emptyset$; $h \leftarrow$ Div; $c \leftarrow 0$
for $t = 1, \dots, T$ **do**
  if $h =$ Div **or** $S_{\text{div}} = \emptyset$ **then**
    $D \leftarrow \text{RANDOMSUBSET}(I, N_g)$
  else
    $D \leftarrow \text{OPENDEPOTS}(S_{\text{div}})$
  end if
  $S \leftarrow \text{ECWA}(D, \kappa_{\text{RCL}})$; $S \leftarrow \text{VND-SO}(S, o)$
  if S is feasible **then**
    if $F_o(S) < F_o(S_{\text{best}})$ **then**
      $S_{\text{best}} \leftarrow S$
    end if
    if $h =$ Div **and** $F_o(S) < F_o(S_{\text{div}})$ **then**
      $S_{\text{div}} \leftarrow S$
    end if
  end if
  $c \leftarrow c + 1$
  if $h =$ Div **and** $c = T_D$ **then**
    $h \leftarrow$ Int; $c \leftarrow 0$
  else if $h =$ Int **and** $c = T_I$ **then**
    $h \leftarrow$ Div; $c \leftarrow 0$; $S_{\text{div}} \leftarrow \emptyset$
  end if
end for
return S_{best}

cost and priority-weighted latency, and Section 5.6 examines how the priority profiles change the service received by individual pharmacies. Section 5.7 reports the robustness and scalability of the heuristic, and Section 5.8 summarizes the managerial implications.

5.1. Experimental Setting

GRASP-VND-SO is implemented in Python 3.9.23, with the route-evaluation routines compiled using Numba 0.60.0. The heuristic experiments were conducted on the Phoenix high-performance computing cluster, equipped with two Intel Xeon E5-2698 v3 processors running at 2.30 GHz (32 physical cores per node), 125 GB of RAM per node, and CentOS Linux 7. Each heuristic run was executed using a single CPU core, while independent runs were parallelized across up to 30 processes. The exact methods were implemented in Python using Gurobi 13.0.2 and executed on an Apple M1 Pro processor with 8 CPU cores, with Gurobi configured to use 8 threads.

Table 1 lists the parameters of GRASP-VND-SO, which are the same for all experiments. Each setting, defined by an instance, an optimization mode, and, for the Tchebycheff mode, a priority profile and a value of λ, is solved by 30 independent runs with seeds $0, \dots, 29$, and the best feasible solution over the 30 runs is reported. The exact models are solved with an absolute optimality tolerance of 10^{-6} and a time limit of 7200 seconds per depot configuration.

Table 1
Parameters of GRASP-VND-SO.

Parameter	Value
GRASP iterations, T	50
RCL size, κ_{RCL}	5
Diversification period, T_D	5
Intensification period, T_I	3
SO adjustment window	4 accepted moves
SO multiplier, ξ	drawn from $\{1.5, 2.5\}$
SO penalty bounds, $[\beta_{\min}, \beta_{\max}]$	$[10^{-6}, 10^{9}]$
Move tolerance, $\varepsilon_{\text{move}}$	10^{-8} (Tchebycheff) 10^{-4} (others)
Tchebycheff augmentation, ρ	10^{-4}
Interior preferences, λ	$\{0.1, 0.2, \dots, 0.9\}$; Isparta: $\{0.1, 0.3, \dots, 0.9\}$
Independent runs per setting	30

5.2. Performance on Standard LLRP Benchmarks

The search is first evaluated on ten standard LLRP benchmark instances, for which results of several published methods are available. In these instances the objective is the total unweighted arrival time, and there are no depot capacities, fixed costs, or route-duration limits; the vehicle capacities, fleet sizes, and numbers of open depots of the original instances are retained. The benchmarks therefore assess the search mechanisms of GRASP-VND-SO rather than the priority-weighted model. For each instance, 30 runs are performed, and the gap between the best run and the best-known solution (BKS) is

$$\text{Gap}_{\text{BKS}}(\%) = 100\,\frac{Z_{\text{best}} - Z_{\text{BKS}}}{Z_{\text{BKS}}}, \tag{49}$$

where Z_{best} is the best latency over the 30 runs and Z_{BKS} the best published value.

5.2.1. Comparison with the Literature

Table 2 compares the best runs of GRASP-VND-SO with the results of RGA and MA (Moshref-Javadi and Lee, 2016), GBILS (Nucamendi-Guillén et al., 2022), the SA-VND variants and M-ILS (Osorio-Mora et al., 2023), and RLHEA (Zou et al., 2024). GRASP-VND-SO attains the BKS on six of the ten instances. On the remaining four, the gaps are 0.05% (Or117-14), 0.88% (Ch100-10), 1.07% (Ch75-10), and 2.63% (Min134-8), and the mean gap over all instances is 0.46%. Even where it does not reach the BKS, GRASP-VND-SO improves on several of the earlier methods, as listed in the last column of the table.

Table 3 summarizes the 30 runs per instance. All runs produced feasible solutions. On the small Gaskell21-5 and Min27-5 instances, every run returned the same solution, whereas the variability is largest on the two biggest instances, Min134-8 and Or117-14, whose mean values exceed the best by 4.9% and 2.0%. The mean CPU time per run grows with instance size, from 8.4 s for Gaskell21-5 to 7,024 s for Min134-8.

Table 2
Best objective values of GRASP-VND-SO and of published methods on the standard LLRP benchmark instances.

Instance	RGA[a]	MA[a]	GBILS[b]	SA-VND$_0^c$	SA-VND$_1^c$	SA-VND$_2^c$	M-ILS[d]	RLHEA[e]	GRASP–VND-SO	BKS Gap (%)	BKS / Outperformed
Ch50-5	1841.70	1690.80	1719.89	**1661.64**	**1661.64**	**1661.64**	**1661.64**	**1661.64**	**1661.64**	**0.00**	*BKS*
Ch75-10	2703.40	2590.30	2399.28	2403.79	2383.04	2400.86	2370.73	**2362.48**	2387.81	1.07	*RGA[a], MA[a], GBILS[b], SA-VND$_0^c$, SA-VND$_2^c$.*
Ch100-10	4116.90	4058.20	3984.05	3791.98	3806.39	3797.00	3803.50	**3788.96**	3822.38	0.88	*RGA[a], MA[a], and GBILS[b].*
Gaskell21-5	699.80	658.40	**653.48**	**653.48**	**653.48**	**653.48**	**653.48**	**653.48**	**653.48**	**0.00**	*BKS*
Gaskell29-5	1238.40	1224.50	**1199.33**	**1199.33**	**1199.33**	**1199.33**	**1199.33**	**1199.33**	**1199.33**	**0.00**	*BKS*
Gaskell32-5	1622.30	1571.00	**1552.84**	**1552.84**	**1552.84**	**1552.84**	**1552.84**	**1552.84**	**1552.84**	**0.00**	*BKS*
Gaskell36-5	1646.10	1642.40	**1627.17**	**1627.17**	**1627.17**	**1627.17**	**1627.17**	**1627.17**	**1627.17**	**0.00**	*BKS*
Min27-5	5387.50†	5387.50†	**5387.55**	**5387.55**	**5387.55**	**5387.55**	**5387.55**	**5387.55**	**5387.55**	**0.00**	*BKS*
Min134-8	25496.00	23387.00	–	21852.35	21910.54	21853.52	**21751.97**	**21751.97**	22324.46	2.63	*RGA[a] and MA[a].*
Or117-14	60580.00	56209.00	–	**53798.53**	53859.08	54103.16	54328.75	**53798.53**	53824.54	0.05	*RGA[a], MA[a], SA-VND$_1^c$, SA-VND$_2^c$, M-ILS[d].*

Notes: Bold values equal the best-known solution (BKS). The BKS gap of GRASP-VND-SO is computed by (49). The last column lists the published methods whose results GRASP-VND-SO improves on. Dashes indicate that no result was reported.
[a] RGA: recursive granular algorithm; MA: memetic algorithm (Moshref-Javadi and Lee, 2016). [b] GBILS: GRASP-based iterated local search (Nucamendi-Guillén et al., 2022). [c] SA-VND$_0$, SA-VND$_1$, SA-VND$_2$: simulated annealing with variable neighborhood descent (Osorio-Mora et al., 2023). [d] M-ILS: multi-start iterated local search (Osorio-Mora et al., 2023). [e] RLHEA: reinforcement-learning-guided hybrid evolutionary algorithm (Zou et al., 2024). † The RGA and MA values for Min27-5 are reported with lower precision than the BKS; the difference of 0.05 is due to rounding.

Table 3
Results of 30 runs of GRASP-VND-SO on the benchmark instances.

Instance	Best	Mean ± SD	CPU (s/run)
Ch50-5	1661.64	1668.05 ± 8.87	126.4
Ch75-10	2387.81	2471.31 ± 43.72	568.7
Ch100-10	3822.38	3907.71 ± 38.27	1640.5
Gaskell21-5	653.48	653.48 ± 0.00	8.4
Gaskell29-5	1199.33	1200.92 ± 1.51	23.7
Gaskell32-5	1552.84	1557.19 ± 9.75	44.5
Gaskell36-5	1627.17	1631.83 ± 4.72	46.8
Min27-5	5387.55	5387.55 ± 0.00	39.6
Min134-8	22324.46	23412.11 ± 540.18	7024.2
Or117-14	53824.54	54915.47 ± 482.62	3269.6

Notes: Best, mean, and standard deviation of the total latency over 30 runs; CPU is the mean time per run. All runs were feasible.

5.2.2. Contribution of the Search Components

The contribution of the main search components is assessed on the same ten instances with four configurations of increasing scope. The baseline, Base, uses five route neighborhoods: relocate, swap, 2-opt/2-opt*, node–arc swap, and arc–arc swap. Base+SO adds the capacity-based strategic oscillation, and Base+SO+DR further adds depot replacement. The final configuration, GRASP-VND-SO, adds the 2-relocate and Swap* neighborhoods and is the method of Section 4. All configurations use the same GRASP settings and 30 runs per instance, without additional parameter tuning; the first three differ by one component at a time, whereas the final step adds two neighborhoods together. Table 4 reports, for each configuration, the mean gap (49) of the best run over the ten instances, the number of instances on which the BKS is matched, the number of instances on which the best run improves on Base, and the mean CPU time per run.

Strategic oscillation alone leaves the mean gap almost unchanged; it improves the best run on three instances but worsens it on two. Adding depot replacement reduces the mean gap from 1.12% to 0.79%. The two additional neighborhoods of the final configuration reduce it further to 0.46%, with four instances improved and none worsened relative to Base, at roughly three times the computational time. The gains are therefore obtained mainly from depot replacement and the extended neighborhood set, whereas strategic oscillation contributes mainly in combination with them. The instance-level results are given in Appendix I.

Table 4
Contribution of the search components on the ten benchmark instances (30 runs per instance).

Configuration	Mean gap (%)	BKS matched	Better than Base	CPU (s/run)
Base	1.13	6	–	393.5
Base+SO	1.12	5	3	464.7
Base+SO+DR	0.79	6	3	465.2
GRASP-VND-SO	0.46	6	4	1,279.2

5.3. Pharmaceutical Instances and Experimental Design

5.3.1. Data and Instance Construction

The case study uses four pharmaceutical distribution networks in Türkiye: Afyon, Aydin, Canakkale, and Isparta. The pharmacies are existing pharmacies of these networks, and their demands are derived from annual sales. The annual sales of pharmacy j are converted into the demand of one distribution shift by assuming 300 operating days per year and five replenishment shifts per day; d_j is therefore measured in shift-level demand units rather than in physical weight or volume. Candidate depots are located at selected pharmacy sites, which serve as proxies for possible locations of distribution capacity rather than existing distribution centers. Road distances and travel durations are given by separate directed matrices obtained from a road network: distances determine the transportation cost, and durations determine arrival times and route durations.

Table 5 reports the characteristics of the four networks. In all networks, every candidate depot has the same opening cost $f_i = 5{,}000$ and capacity W, and every vehicle has

Table 5
Characteristics of the four pharmaceutical distribution networks.

City	n	m	N_g	N_v	D_{tot}	$d_{\max}$	Diam. (km)	L (s)	W	Q
Afyon	14	5	2	4	735	136	17.87	10,000	435	280
Aydin	31	5	3	5	621	40	145.55	18,900	245	140
Canakkale	37	5	2	5	545	29	8.22	10,000	325	125
Isparta	48	5	2	6	400	60	9.79	10,000	240	75

Notes: n: number of pharmacies; m: number of candidate depots; N_g: number of open depots; N_v: maximum number of vehicles; D_{tot}: total demand; $d_{\max}$: largest pharmacy demand; Diam.: geographical diameter of the network; L: maximum route duration; W: depot capacity; Q: vehicle capacity. Demands and capacities are in shift-level demand units.

the deployment cost $F_i = 200$. The travel cost c_{uv} of an arc equals its road distance in kilometers. The operational parameters were calibrated according to the shift-level demand and geographical characteristics of each network. The number of open depots N_g was determined by the candidate-depot configuration and network dispersion. Depot capacity W was obtained by rounding $D_{\text{tot}}/(0.85N_g)$ upward to the nearest five demand units, targeting approximately 85% depot utilization. The initial fleet size N_v was based on a target of approximately nine pharmacies per route, while vehicle capacity Q was selected to achieve approximately 89% fleet utilization, subject to individual pharmacy demands. The fleet configurations were adjusted where necessary to ensure sufficient capacity across the open depots, particularly in Afyon and Aydin. The maximum route duration L was selected considering the geographical extent and road-travel times of each network, with 10,000 seconds for Afyon, Canakkale and Isparta and 18,900 seconds for the more geographically dispersed Aydin network. Feasibility was assessed through depot and fleet capacity checks and confirmed by obtaining feasible routing solutions satisfying customer coverage, depot and vehicle capacities, fleet availability, and route-duration constraints. These parameters are kept fixed in all experiments, so that differences between solutions are due to the priority profiles and the trade-off preferences only.

5.3.2. Priority Profiles and Preferences

Five priority profiles are considered. They represent alternative service policies rather than clinically validated urgency classes, and the same coefficients are used in all runs of a profile.

- Equal (E): $p_j = 1$ for all $j \in J$.
- Tiered (T): about 20% of the pharmacies receive $p_j = 1.0$, 40% receive $p_j = 0.7$, and the remaining pharmacies receive $p_j = 0.4$.
- Demand-correlated (DC): $p_j = 0.5 + 0.5\,(d_j - d_{\min})/(d_{\max} - d_{\min})$, so that priorities increase with demand.
- Negative-correlated (NC): $p_j = 1 - 0.5\,(d_j - d_{\min})/(d_{\max} - d_{\min})$, so that priorities decrease with demand.
- Hyper-critical (HC): two predetermined pharmacies receive $p_j = 10$ and all others $p_j = 1$.

The Tiered and Hyper-critical priority assignments were generated using fixed random seeds. In the Tiered profile, pharmacies were independently assigned to the three priority classes with probabilities of 0.20, 0.40, and 0.40. In the Hyper-critical profile, two pharmacies were randomly selected without replacement. These assignments remained fixed across all optimization runs and preference settings for each network.

For Afyon, Aydin and Canakkale, the two anchors and the nine interior preferences $\lambda \in \{0.1, 0.2, \ldots, 0.9\}$ are solved for every profile; for Isparta, whose runs take considerably longer (Section 5.7), the five preferences $\lambda \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ are solved. To compare the networks on a common basis, Sections 5.5–5.7 report these five preferences, which gives seven solutions per setting, whereas all nine are used in the exact validation on Afyon in Section 5.4.2. The anchors are obtained from 30 runs in the financial and latency modes and selected lexicographically over the runs: the cost anchor is the run with the smallest financial cost and, among ties, the smallest latency, and the latency anchor is defined analogously. Since the financial cost does not depend on the priorities, the cost runs are shared by all profiles of a network, and their latencies are recomputed for each profile before the selection. The heuristic thus defines its own normalization, and the exact anchors of Section 5.4.1 are used only to evaluate it. The anchor values of all settings are listed in Appendix J. Because different profiles weight the arrival times differently, priority-weighted latencies are compared only within a profile; comparisons across profiles use the common evaluations of Section 5.6.

5.4. Validation against Exact Solutions

The heuristic solutions are compared with exact solutions at two levels. For Afyon, the anchors and all interior preferences are solved exactly, whereas for the larger networks the exact methods are applied to the anchor problems only.

Each anchor of (37)–(38) is obtained by two problems solved in sequence. For the cost anchor, C_F is minimized first, and P is then minimized subject to $C_F(S) \le \widehat{C}_{\min} + 10^{-6}$; the latency anchor is obtained in the same way with the roles of the two criteria exchanged. These problems and the interior problems (41) of Afyon are solved with the commodity-flow formulation of Section 3.4, in which the

latency is evaluated by the priority flow P^F and the arrival-time windows, preprocessing rules and fleet inequality are included. This formulation is effective for the cost problems, but its bounds for problems involving the latency weaken quickly as the number of pharmacies grows. For the anchor problems that it does not solve within the time limit, in particular the latency anchors of the larger networks, a branch-and-price algorithm is used, which solves a route-based formulation for each set of N_g open depots and is described in Appendix D.

An exact value is reported as certified if optimality is proven within an absolute tolerance of 10^{-6}. Otherwise, the best feasible value UB and the best lower bound LB are reported together with the gap $\text{Gap}_{\text{exact}} = 100\,(UB - LB)/UB$. A heuristic value z_H is compared with a certified optimal value z^* through the deviation $\text{Dev}_H = 100\,(z_H - z^*)/z^*$; a zero deviation means that the heuristic attains the optimal objective value, although its routes may differ from those of the exact solution.

5.4.1. Anchor Values

Table 6 reports the exact anchors of the three networks for which they were computed, together with the deviation of the heuristic first-stage anchors, that is, the best of 30 runs in the financial and latency modes. For Afyon, all 20 anchor values are certified, and the heuristic attains each of them; each hierarchical problem was solved in 4.9 to 21.0 s. For Canakkale, 19 of the 20 anchor values are certified; the exception is the latency of the cost anchor under the Hyper-critical profile, whose second stage reached the time limit. The heuristic again attains all ten first-stage anchors exactly, so that on a network of 37 pharmacies GRASP-VND-SO finds the optimal cost and latency anchors of every profile. In addition, the explicit second stages of Canakkale coincide with the values obtained by evaluating the first-stage solutions, which indicates that the cost-optimal and latency-optimal solutions are unique there.

Aydin is the most difficult network. Its cost anchor is certified, and the heuristic cost anchor exceeds it by 0.10% of the financial cost, which corresponds to 2.2% of the travel cost. The latency anchors are certified for the Tiered and Demand-correlated profiles; for the Equal and Hyper-critical profiles, branch-and-price leaves gaps of 0.14% and 0.91%. Under the Negative-correlated profile, the best heuristic solution is better than the branch-and-price incumbent and is within 0.19% of its lower bound. For the other four profiles, the heuristic latency anchors exceed the best-known values by 0.67% to 5.33%. The latency mode of the heuristic is therefore less reliable on this network, which has the largest diameter and three open depots. Because the heuristic cost anchor of Aydin is not optimal, several interior solutions of this network are cheaper than the anchor, which affects its preference paths (Section 5.5). The second-stage values $\widehat{P}_C$ and $\widehat{C}_P$ of Aydin in Table 6 are obtained by evaluating the first-stage solutions; since not all first stages are certified, they are upper bounds on the lexicographic values.

5.4.2. Interior Preferences on Afyon

Because the Afyon anchors are certified, the scalarized problems of this network are fully defined, and the 45 interior problems (five profiles and nine values of λ) were solved exactly with the scalarized model (41), using the priority-flow latency. All 45 problems were solved to proven optimality, in 3.7 to 32.2 s each, and every optimal solution was verified independently from its routes. At each of the 45 preferences, the best of the 30 GRASP-VND-SO runs attains the certified optimal value of Z_λ, with the same financial cost and priority-weighted latency; in some cases the heuristic and exact solutions differ in their routes but not in their objective values. Table 7 summarizes the results by profile. On this small network, a single exact solve is faster than a heuristic run, which takes 34 to 44 s on average; Section 5.4.3 shows that this advantage disappears quickly as the networks grow.

5.4.3. Limits of Exact Solution

The exact methods scale very differently with the objective. Problems in which only the financial cost is minimized are solved quickly, because the fixed depot and vehicle charges dominate the objective and only the travel costs have to be optimized: with the commodity-flow formulation, the cost anchor of Aydin ($n = 31$) is certified in 41 minutes. Problems involving the latency are much harder. Branch-and-price certifies the latency anchors of Canakkale in 15 to 75 minutes for four profiles, whereas for the Hyper-critical profile a single depot pair required about 6.5 hours; on Aydin, it certifies two latency anchors within the limit of two hours per depot set and leaves gaps of up to 0.91% for the other three. The interior Tchebycheff problems combine both criteria and are harder still: after 30 minutes, the commodity-flow formulation of the problem with $\lambda = 0.5$ under the Equal profile still has a gap of 98.5% on Canakkale and of 229% on Aydin. In normalized form, the linear relaxation bounds are weak because the gap of the relaxation exceeds the range Δ_C of the financial cost. Exact certification of interior preferences is therefore limited to the smallest network, and GRASP-VND-SO is used for the larger ones.

5.5. Cost-Latency Trade-offs

In every network, the number of open depots is fixed at N_g, and the vehicle capacity forces all solutions to use the N_v available vehicles.

The depot and vehicle charges are therefore constant across all solutions of a network and constitute the dominant share of C_F. Relative to the cost anchor, changes in C_F remain below 0.35% for all interior preferences, although larger increases occur at some latency anchors, reaching 1.44% for the Aydin Hyper-critical profile. Consequently, changes in C_F provide limited insight into the underlying routing trade-off.

The analysis therefore uses the travel cost $C_T(S) = C_F(S) - \sum_{i\in I} f_i y_i - \sum_{i\in I} F_i k_i$, which equals the distance travelled in kilometers. For the solution S_λ obtained with preference λ, the increase in travel cost and the reduction in

Table 6
Exact anchors and deviation of the heuristic first-stage anchors.

		Cost anchor		Latency anchor			Heuristic dev. (%)	
City	**Profile**	$\widehat{C}_{\min}$	$\widehat{P}_C$	$\widehat{P}_{\min}$	$\widehat{C}_P$	**Gap (%)**	$\widehat{C}_{\min}$	$\widehat{P}_{\min}$
Afyon	Equal	10,859.963	18,594	7,791	10,896.869	0.00	0.00	0.00
	Tiered	10,859.963	11,864.7	4,298.7	10,874.510	0.00	0.00	0.00
	Demand-correlated	10,859.963	10,211.67	4,317.85	10,877.462	0.00	0.00	0.00
	Negative-correlated	10,859.963	17,679.33	6,856.28	10,868.306	0.00	0.00	0.00
	Hyper-critical	10,859.963	26,964	12,641	10,873.529	0.00	0.00	0.00
Canakkale	Equal	11,031.218	20,505	16,511	11,062.137	0.00	0.00	0.00
	Tiered	11,031.218	12,513.3	10,158.7	11,041.097	0.00	0.00	0.00
	Demand-correlated	11,031.218	13,265.68	11,083.58	11,064.790	0.00	0.00	0.00
	Negative-correlated	11,031.218	17,491.83	13,635.25	11,055.721	0.00	0.00	0.00
	Hyper-critical	11,031.218	28,236[a]	17,251	11,062.996	0.00	0.00	0.00
Aydin	Equal	16,741.297	150,827[b]	95,809	16,922.477[b]	0.14	0.10	0.67
	Tiered	16,741.297	79,697.9[b]	52,436.9	16,930.136[b]	0.00	0.10	1.17
	Demand-correlated	16,741.297	91,590.96[b]	58,487.15	16,922.477[b]	0.00	0.10	3.37
	Negative-correlated	16,741.297	134,649.54[b]	84,221.13[c]	16,849.745[b]	0.19	0.10	0.00
	Hyper-critical	16,741.297	194,378[b]	140,092	16,981.612[b]	0.91	0.10	5.33

Notes: Exact values from the commodity-flow formulation (cost anchors; all anchors of Afyon) and branch-and-price (latency anchors of Canakkale and Aydin). Gap: optimality gap of $\widehat{P}_{\min}$; all other first-stage values are certified. Heuristic dev.: deviation Dev_H of the best of 30 heuristic runs from the best known value. [a]Upper bound; the second stage reached the time limit. [b]Evaluation of the first-stage solution (upper bound on the lexicographic value). [c]Heuristic solution, better than the branch-and-price incumbent (84,313.37). Isparta: no exact values computed.

Table 7
Exact validation of the interior preferences on Afyon.

Profile	At optimum	Distinct points	Exact (s)	Heuristic (s)
Equal	9/9	8	8.8	33.7
Tiered	9/9	9	17.6	42.0
Demand-correlated	9/9	8	8.2	41.9
Negative-correlated	9/9	7	15.8	41.6
Hyper-critical	9/9	8	12.0	44.1

Notes: At optimum: number of preferences at which the best heuristic run attains the certified optimal Z_λ. Distinct points: number of different Pareto points obtained for the nine values of λ. Exact: mean time of an exact solve; Heuristic: mean time of the best of the 30 runs.

priority-weighted latency relative to the cost anchor S^C are

$$\Delta C_T(\lambda) = 100\,\frac{C_T(S_\lambda) - C_T(S^C)}{C_T(S^C)}, \tag{50}$$

$$\Delta P(\lambda) = 100\,\frac{P(S^C) - P(S_\lambda)}{P(S^C)}. \tag{51}$$

Both measures are computed within a profile, so that they can be compared across profiles even though the latencies themselves cannot.

Table 8 reports the two measures for the five interior preferences and the latency anchor (the corresponding absolute values are given in Appendix K), and Figure 1 shows the corresponding approximated Pareto fronts in terms of travel cost and priority-weighted latency. In the 15 settings of Afyon, Canakkale, and Isparta, the seven solutions of a path are distinct and mutually nondominated, and the travel cost increases and the latency decreases monotonically with λ. Several of these solutions could not have been obtained by a weighted-sum scalarization. A solution that lies strictly above the segment joining two other solutions of its front is unsupported, because for every weighting of C_F and P one of these two solutions has a smaller weighted sum. On Afyon, where the interior solutions are certified optimal, 9 of the 25 interior solutions of the five profiles are unsupported Pareto-optimal solutions. On Canakkale and Isparta, 8 and 2 of the 25 interior solutions lie above the segment joining two other solutions of their fronts. These solutions illustrate the advantage of the Tchebycheff scalarization discussed in Section 3.5.

The paths of Aydin show the effect of a heuristic anchor that is not optimal. Nine of its 25 interior solutions are cheaper than the cost anchor, so that their travel-cost changes in Table 8 are negative and their normalized cost $\overline{C}$ is slightly below zero; under the Equal profile, for example, the solutions for $\lambda \in \{0.1, 0.3, 0.5\}$ coincide and dominate the cost anchor. Under the Hyper-critical profile, the solution for $\lambda = 0.9$ also has a slightly lower latency than the latency anchor. Over the five profiles, 11 of the 35 solutions are dominated by another solution of the same path, and the paths are not monotone in λ. These solutions are reported as obtained, and the dominated ones are excluded from the approximated fronts in Figure 1. The Tchebycheff properties of Section 3.5 are not affected, but the example shows that the quality of the anchors determines how evenly a given set of preferences covers the front.

The paths are strongly concave, so that most of the attainable latency reduction is obtained at a small increase in travel cost. At $\lambda = 0.5$, the latency is reduced by 45.2–52.8% in Afyon at an increase in travel cost of 3.4–5.7%, by 28.6–35.5% in Isparta at 9.5–11.9%, by 12.9–32.0% in Canakkale at 10.2–18.4%, and by 24.8–34.7% in Aydin at changes in travel cost between −0.4% and 4.6%. The last part of the latency reduction is expensive. In Isparta, moving from $\lambda = 0.9$ to the latency anchor reduces the latency by a further 1.8 to 2.1 percentage points, whereas the increase in travel cost grows from 17.3–21.0% to 36.8–50.0%. In Canakkale, the Equal profile already requires 73.1% more travel at $\lambda = 0.9$ for a latency reduction of 17.9%, and the latency anchor requires 99.0% for 19.5%. The latency anchors are therefore poor operating points, and their role is mainly to fix the normalization of Section 3.5.

Under the Equal profile, $P(S)/|J|$ is the mean arrival time at the pharmacies, which gives the trade-off a direct operational meaning. At $\lambda = 0.5$, the mean arrival time falls from 22.1 to 10.4 minutes in Afyon for 3.4 km of additional travel per shift, from 17.1 to 11.7 minutes in Isparta for 6.9 km, and from 9.2 to 7.8 minutes in Canakkale for 5.6 km. In Aydin, the mean arrival time falls from 75.5 to 55.9 minutes, and the travel cost even decreases by 2.8 km, because the heuristic cost anchor is not optimal. Canakkale is the most compact network, and the arrival times of its cost anchor are already short, so that its latency can be reduced only by about 16–22% under the first four profiles. The Hyper-critical profile is the exception, with a reduction of 38.9% at the latency anchor, because the two pharmacies with the large weight can be moved to the first positions of their routes.

The preference also affects the location decisions. In Afyon, Canakkale and Isparta, the set of open depots changes along some of the paths; for example, under the Hyper-critical profile, Isparta opens depots D02 and D04 up to $\lambda = 0.5$ and depots D01 and D02 from $\lambda = 0.7$ on, and Canakkale opens D02 and D05 instead of D02 and D03 for $\lambda \in \{0.3, 0.5, 0.7\}$. The service criterion therefore influences where the depots are opened, which would be missed by an approach that fixes the depots before the routes are designed.

5.6. Effect of Priority Policies

The priority-weighted latencies of different profiles cannot be compared, because each profile weights the arrival times differently. The profiles are therefore compared by measures that do not depend on the priority coefficients: the mean arrival time $\overline{T}(S) = \frac{1}{|J|}\sum_{j\in J} a_j(S)$, the latest arrival time $T_{\max}(S) = \max_{j\in J} a_j(S)$, and the mean arrival time $\overline{T}_{\mathrm{HC}}(S) = \frac{1}{2}\sum_{j\in J_{\mathrm{HC}}} a_j(S)$ of the set J_{HC} of the two pharmacies that receive $p_j = 10$ in the Hyper-critical profile. The arrival times are recomputed from the routes of each solution with the duration matrix; for the Equal profile, the recomputed latencies coincide with the recorded ones. Because each profile is normalized with its own anchors, the same value of λ does not correspond to the same cost. At $\lambda = 0.5$, however, the travel costs of the five profiles differ by at most 1.4 km in Afyon, 2.6 km in Canakkale and 1.6 km in Isparta, so that the solutions at this preference are compared at approximately equal cost. In Aydin, the differences reach 37.9 km, or 5% of the travel cost, and the comparison is less tight. Table 9 reports these solutions.

The Hyper-critical profile has a large and targeted effect. In Canakkale, its two critical pharmacies are reached after 0.6 minutes on average, compared with 8.7 to 14.0 minutes under the other profiles, and in Isparta after 8.2 minutes, compared with 25.4 to 27.6 minutes. The cost of this improvement for the other pharmacies is small: the mean arrival time increases by 0.6 minutes (7.6%) in Canakkale and by 0.8 minutes (6.9%) in Isparta relative to the Equal profile. Aydin, the most dispersed network, shows the same effect on a larger scale: the critical pharmacies are reached after 33.2 instead of 120.6 to 123.0 minutes, but the mean arrival time of all pharmacies increases by 12.9 minutes (23%) and the travel cost by 5%. The effect persists over the whole path: for the five interior preferences, $\overline{T}_{\mathrm{HC}}$ lies between 0.6 and 6.6 minutes in Canakkale and between 4.8 and 9.0 minutes in Isparta under the Hyper-critical profile, compared with 0.6 to 16.0 and 17.2 to 27.6 minutes under the Equal profile, which reaches the critical pharmacies early only incidentally. In Canakkale, the Hyper-critical profile also changes the location decision at $\lambda = 0.5$ and opens depot D05 instead of D03. In Afyon, the two critical pharmacies are close to an open depot and, at $\lambda = 0.5$, are reached within about five minutes under all profiles except the Tiered one; the Hyper-critical profile makes a difference mainly at small values of λ, where it reduces $\overline{T}_{\mathrm{HC}}$ from 21.7 minutes under the Equal profile to 6.5 minutes at $\lambda = 0.1$.

The Tiered, Demand-correlated, and Negative-correlated profiles spread the weights over many pharmacies and change the mean arrival time only slightly. At $\lambda = 0.5$, their mean arrival times exceed that of the Equal profile by at most 0.3 minutes in Canakkale and 0.5 minutes in Isparta. In Afyon, the Negative-correlated profile, which gives low priority to the pharmacies with large demand, increases the mean arrival time by 1.8 minutes and the latest arrival from 27.1 to 42.9 minutes.

For the pharmacies targeted by each profile, the mean arrival times at $\lambda = 0.5$ are compared with those of the same pharmacies under the Equal profile. These are the pharmacies with $p_j = 1.0$ in the Tiered profile and the 20% with the largest p_j (including ties) in the Demand-correlated and Negative-correlated profiles. Under the Tiered profile, they are reached 0.7–2.3 minutes earlier in all four networks. Under the Demand-correlated profile, they are reached 3.0 minutes earlier in Canakkale but 1.5 minutes later in Isparta, with differences of at most 0.5 minutes in Afyon and Aydin. Under the Negative-correlated profile, they are reached 5.3 minutes later in Afyon, which may partly reflect the lower travel cost of this solution (Table 9), and differ by less than 0.6 minutes elsewhere. A lower priority-weighted latency therefore does not necessarily shorten the arrival times of every high-priority subgroup.

Table 8
Increase in travel cost and reduction in priority-weighted latency along the preference paths, relative to the cost anchor (%).

		Cost anchor		$\lambda = 0.1$		$\lambda = 0.3$		$\lambda = 0.5$		$\lambda = 0.7$		$\lambda = 0.9$		Latency anchor	
City	**Profile**	C_T	P	ΔC_T	ΔP	ΔC_T	ΔP	ΔC_T	ΔP	ΔC_T	ΔP	ΔC_T	ΔP	ΔC_T	ΔP
Afyon	Equal	59.963	18,594	2.6	34.3	3.8	51.1	5.7	52.8	7.3	56.1	11.0	57.0	61.5	58.1
	Tiered	59.963	11,864.7	1.8	16.7	3.4	45.6	4.4	50.9	7.3	58.8	11.9	60.0	24.3	63.8
	Demand-correlated	59.963	10,211.7	2.2	22.6	3.4	41.3	5.7	48.4	7.3	51.4	16.2	55.3	29.2	57.7
	Negative-correlated	59.963	17,679.3	1.2	12.9	2.6	38.1	3.4	46.6	3.8	53.7	7.3	58.8	13.9	61.2
	Hyper-critical	59.963	26,964	1.2	13.7	3.1	36.6	3.8	45.2	5.7	49.2	9.0	50.5	22.6	53.1
Aydin	Equal	757.591	140,352	−0.4	25.9	−0.4	25.9	−0.4	25.9	4.4	27.2	6.9	29.3	12.2	31.3
	Tiered	757.591	79,756.8	0.6	19.1	−0.4	27.5	2.0	27.5	4.6	29.3	7.6	32.8	15.0	33.5
	Demand-correlated	757.591	88,214.0	−0.4	27.4	1.0	23.1	−0.2	27.4	6.4	24.6	6.1	28.9	12.2	31.5
	Negative-correlated	757.591	122,314.0	−0.4	24.8	−0.4	24.8	−0.4	24.8	3.8	24.5	6.2	28.2	12.2	31.1
	Hyper-critical	757.591	250,944	0.6	34.4	1.0	34.7	4.6	34.7	6.3	38.1	4.4	41.2	31.9	41.2
Canakkale	Equal	31.218	20,505	4.2	11.3	9.8	14.7	17.9	15.9	26.0	17.2	73.1	17.9	99.0	19.5
	Tiered	31.218	12,513.3	2.5	5.8	5.4	11.1	10.2	12.9	14.7	15.1	18.9	17.7	31.6	18.8
	Demand-correlated	31.218	13,265.7	4.8	10.3	10.5	12.3	15.7	14.3	18.3	15.4	23.2	16.1	107.5	16.4
	Negative-correlated	31.218	17,491.8	4.1	11.9	9.7	16.2	15.2	17.7	21.4	19.1	64.8	20.0	78.5	22.0
	Hyper-critical	31.218	28,236	8.1	10.0	14.6	25.8	18.4	32.0	27.9	33.9	56.8	36.7	101.8	38.9
Isparta	Equal	63.775	49,168	3.8	12.7	8.2	25.9	10.8	31.7	14.4	35.4	20.9	38.6	50.0	40.4
	Tiered	63.775	32,172.7	3.6	15.8	7.3	29.1	10.7	35.5	15.0	38.7	21.0	42.3	50.0	44.4
	Demand-correlated	63.775	26,742.6	2.6	13.8	6.4	23.0	9.5	28.6	13.8	32.6	17.3	37.0	36.8	38.9
	Negative-correlated	63.775	47,009.4	2.3	18.1	6.4	26.5	9.5	31.9	12.5	35.8	19.2	39.3	40.3	41.4
	Hyper-critical	63.775	67,969	3.3	16.1	7.5	28.1	11.9	34.2	15.7	39.6	20.0	44.3	46.1	46.1

Notes: C_T: travel cost (km) and P: priority-weighted latency (s) of the cost anchor. ΔC_T and ΔP: Eqs. (50)–(51). Each interior entry is the best of 30 runs. Latencies refer to the priority vector of the profile and are not comparable across profiles.

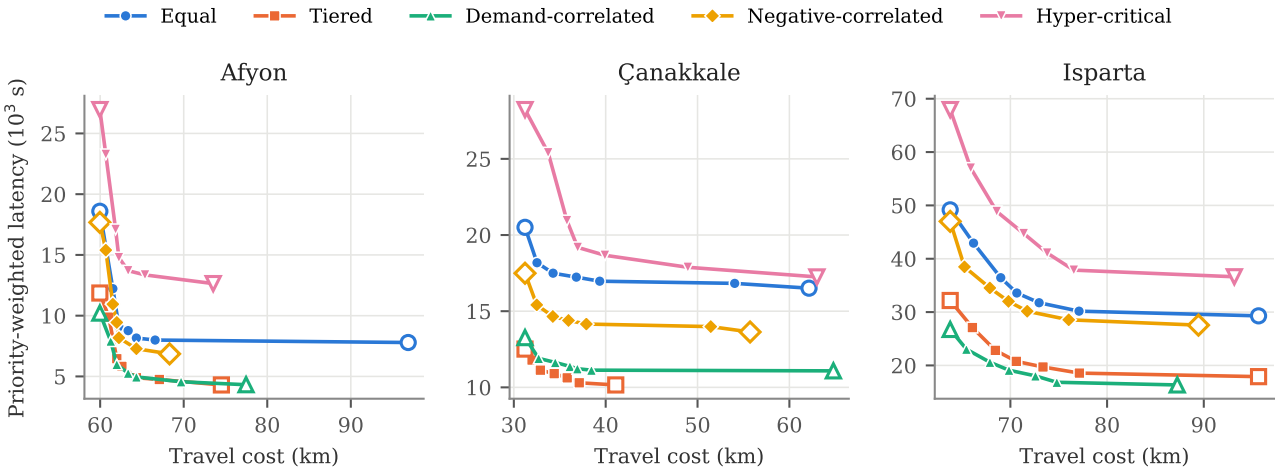


Figure 1: Approximated Pareto fronts of the five priority profiles: priority-weighted latency against travel cost. Each front connects the cost anchor, the solutions for $\lambda \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$, and the latency anchor; the anchors are shown by open symbols, and dominated solutions by crosses. Each latency is measured with the priority vector of its own profile, so that the vertical positions of different profiles are not comparable.

5.7. Robustness and Scalability of the Heuristic

All 100 city–profile–preference settings achieve a feasibility rate of 100%, but the repeatability of the best value differs across networks. The mean $\overline{Z}_\lambda$ exceeds Z^*_λ by at most 0.010 in Afyon, 0.034 in Canakkale, and 0.035 in Isparta, but by 0.06 to 0.51 in Aydin. The best value is attained by 5 to 30 runs in Afyon and by 1 to 30 runs in Canakkale, but by at most three runs in Aydin and by a single run in every Isparta setting. The large dispersion in Aydin is consistent with its large diameter, since run-to-run differences in route length are large relative to its cost range ΔC (Table J.1), and with the lower reliability of the heuristic on this network noted in Section 5.4.1. GRASP-VND-SO therefore consistently constructs feasible solutions, while the repeatability of the best value depends on the network and the priority profile.

On Afyon, the best run attains the certified optimum of Section 5.4.2 at every preference. On Isparta, under the Equal profile, the average run at $\lambda = 0.5$ has a 2.0% higher latency and a 1.6% higher travel cost than the best run. Taking the best of several independent runs is therefore useful on the larger and more dispersed instances.

Table 11 shows how computing time increases with network size. The mean computing time per run increases from 39.4 s for Afyon ($n = 14$) to 667.7 s for Aydin ($n =$

Table 9
Common evaluation of the solutions of the five profiles at $\lambda = 0.5$.

City	Profile	Depots	C_T	$\overline{T}$	$\overline{T}_{\mathrm{HC}}$	$T_{\max}$
Afyon	Equal	D01,D04	63.4	10.4	4.6	27.1
	Tiered	D01,D04	62.6	11.2	13.0	27.1
	Demand-correlated	D01,D04	63.4	10.4	4.6	27.1
	Negative-correlated	D01,D04	62.0	12.3	5.3	42.9
	Hyper-critical	D01,D04	62.2	10.8	5.3	27.1
Aydin	Equal	D01,D02,D03	754.8	55.9	120.6	191.9
	Tiered	D01,D02,D03	772.6	56.0	123.0	196.8
	Demand-correlated	D01,D02,D03	756.3	56.0	120.6	191.9
	Negative-correlated	D01,D02,D03	754.8	55.9	120.6	191.9
	Hyper-critical	D01,D02,D03	792.7	68.8	33.2	225.4
Canakkale	Equal	D02,D03	36.8	7.8	12.4	19.2
	Tiered	D02,D03	34.4	8.0	9.9	19.2
	Demand-correlated	D02,D03	36.1	8.1	8.7	21.0
	Negative-correlated	D02,D03	36.0	7.9	14.0	19.6
	Hyper-critical	D02,D05	37.0	8.3	0.6	18.4
Isparta	Equal	D02,D04	70.7	11.7	27.6	42.6
	Tiered	D02,D04	70.6	12.1	25.4	40.0
	Demand-correlated	D02,D04	69.9	11.9	27.4	38.8
	Negative-correlated	D02,D04	69.8	11.8	27.5	39.1
	Hyper-critical	D02,D04	71.4	12.5	8.2	39.6

Notes: Depots: open depots. C_T: travel cost (km). $\overline{T}$, $\overline{T}_{\mathrm{HC}}$ and $T_{\max}$: mean arrival time, mean arrival time of the two Hyper-critical pharmacies, and latest arrival time (min).

Table 10
Robustness of GRASP-VND-SO over 30 runs (Afyon, Equal-Priority profile).

λ	FR (%)	Z^*_λ	$\overline{Z}_\lambda$	$s_{Z,\lambda}$	Hits	Time (s)
0.1	100	0.0410	0.0479	0.0071	12	31.4
0.3	100	0.0432	0.0444	0.0043	28	33.2
0.5	100	0.0459	0.0459	0.0000	30	36.5
0.7	100	0.0356	0.0356	0.0000	30	34.6
0.9	100	0.0179	0.0191	0.0019	21	34.4

Notes: FR: percentage of feasible runs. Hits: runs attaining the best recorded Z^*_λ within a tolerance of 10^{-9}. The mean and sample standard deviation are computed over feasible runs. Time: mean recorded runtime of a run with 50 GRASP iterations.

31), 1,750.5 s for Canakkale ($n = 37$), and 5,291.5 s for Isparta ($n = 48$). A log-log regression across these four networks yields an empirical scaling exponent of approximately 3.94 ($R^2 = 0.9945$). This empirical relationship reflects the computational demands of the neighborhood searches performed within the 50 GRASP iterations, rather than a theoretical complexity bound. The independent runs are executed in parallel, so that the 150 runs of an Isparta profile required a wall-clock time of about 13.7 hours on 16 cores. The computing time of individual runs therefore limits the size of the networks that can be solved with the present settings; larger networks would require fewer GRASP iterations, restricted neighborhood evaluations, or parallel neighborhood exploration.

Table 11
Computing time of a GRASP-VND-SO run by network.

City	n	m	Mean (s)	Min (s)	Max (s)
Afyon	14	5	39.4	26.0	52.7
Aydin	31	5	667.7	471.2	899.1
Canakkale	37	5	1,750.5	1,152.9	2,898.6
Isparta	48	5	5,291.5	3,905.5	7,528.5

Notes: Mean, minimum, and maximum elapsed times are calculated over all 750 independent runs per network, covering five priority profiles, five preference values ($\lambda = 0.1, 0.3, 0.5, 0.7, 0.9$), and 30 seeds per setting. Each run uses 50 GRASP iterations.

5.8. Managerial Implications

The results suggest three implications for the planning of pharmaceutical distribution. First, a substantial improvement in service can be obtained at a small cost. In all four networks, the fixed depot and vehicle charges are the same for all solutions, and service is improved by changing the assignment of the pharmacies to depots and vehicles and the order of the visits, at the price of slightly longer routes. Under the Equal profile at $\lambda = 0.5$, the mean arrival time falls from 22.1 to 10.4 minutes in Afyon for 3.4 km of additional travel per shift, and from 17.1 to 11.7 minutes in Isparta for 6.9 km. The fronts are strongly concave, so that a planner gains little by moving toward the latency anchor: in Isparta, the last 1.8 to 2.1 percentage points of latency reduction increase the travel cost by a further 19.5 to 29.1 percentage points. A small number of preferences around the knee of the front, for example $\lambda \in \{0.3, 0.5\}$, is therefore sufficient to identify the operating points of practical interest.

Second, the effect of a priority policy depends on how the weights are distributed rather than on the profile itself. Concentrating a large weight on a few pharmacies, as in the Hyper-critical profile, moves them to the beginning of their routes: in Canakkale they are reached after 0.6 instead of 8.7 to 14.0 minutes and in Isparta after 8.2 instead of 25.4 to 27.6 minutes, whereas the mean arrival time of all pharmacies increases by less than one minute. Moderate weights spread over many pharmacies, as in the Tiered and correlated profiles, change the service of the network only slightly. A distributor who needs to guarantee early delivery to a few pharmacies, such as those serving hospitals or emergency services, should therefore give them clearly larger weights than the others.

Third, the location of the depots should be decided together with the service policy. The set of open depots changes with the preference in Afyon, Canakkale, and Isparta and with the priority profile in Canakkale, where the Hyper-critical profile opens a different depot from the other profiles. A sequential approach that fixes the depots on the basis of cost and designs the routes afterward would not find these solutions.

6. Conclusion

This study presents a bi-objective priority-weighted LLRP that integrates financial distribution cost and pharmacy-priority-weighted cumulative arrival time with depot location, customer assignment, fleet utilization and route sequencing. Anchor-normalized augmented weighted Tchebycheff scalarization represents alternative cost-service preferences without introducing a monetary delay penalty and, unlike a weighted sum, can reach unsupported Pareto-optimal solutions. The theoretical properties establish Pareto efficiency for globally optimal interior scalarizations and invariance under a common scaling of the priorities; these properties do not certify heuristic solutions. For exact reference solutions, a commodity-flow reformulation represents the priority-weighted latency by a flow of priority weights, and a branch-and-price algorithm solves the harder anchor problems.

The GRASP-VND-SO heuristic combines randomized ECWA construction, seven route-level neighborhoods, depot replacement and capacity-based strategic oscillation. On ten standard LLRP benchmarks, it attains the best-known solution on six instances with a mean gap of 0.46%. On the pharmaceutical networks, its best run attains all anchors of Afyon, all first-stage anchors of Canakkale and the certified optimum at all 45 interior preferences of Afyon. It is less reliable on Aydin, the most dispersed network, where its anchors deviate from the best known values by up to 5.33%.

The case study of four pharmaceutical distribution networks in Türkiye under five priority profiles yields three main insights. First the fronts are strongly concave: at $\lambda = 0.5$, the priority-weighted latency is 12.9–52.8% lower than at the cost anchor, with travel cost rising by at most 18.4%. Second, the effect of a priority policy depends on how its weights are distributed. Concentrated weights, as in the Hyper-critical profile, move the critical pharmacies to the beginning of their routes at a small cost to the others, whereas weights spread over many pharmacies change the service only slightly and do not necessarily shorten the arrival times of every high-priority subgroup. Third, the set of open depots changes with both the preference and the priority profile, so depot location and service policy should be decided together.

The study has several limitations. The input data are deterministic and refer to a single distribution shift, service times and delivery deadlines are not modeled, and the priority profiles are illustrative service policies rather than clinically established urgency classes. In the case studies, the number of open depots is fixed and all available vehicles are used, so the trade-off is expressed through travel cost only. Except for the interior preferences of Afyon, the reported fronts are heuristic approximations of a representative subset of the Pareto front. The computing time of GRASP-VND-SO limits the size of the networks that can be solved with the present settings. Future research could strengthen the formulation of the scalarized problems, whose relaxations remain weak on the larger networks, so that interior trade-off points can be certified beyond the smallest network. The model could also be extended to uncertain demand and travel times, time-dependent travel conditions, dynamic priorities and variable numbers of depots and vehicles.

Data Availability

The operational data used to construct the pharmaceutical case-study instances are subject to confidentiality restrictions and are therefore not deposited in a public repository. Aggregate anchor references for Afyon, Aydin, Canakkale and Isparta, together with anonymized production records for the completed Afyon, Aydin, and Canakkale priority profiles have been organized as machine-readable supplementary files, including selected objective pairs, run-level outcomes, and an explicit production-coverage index; public release remains subject to author review of confidentiality and repository requirements. Upon reasonable request, the authors may provide additional anonymized pharmacy-demand data, network inputs, and parameter settings, subject to applicable data-sharing restrictions.

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A. Notation and Parameters

Table A.1 summarizes the sets, parameters, decision variables, and solution measures used throughout the paper.

B. Abbreviations

Table B.1 lists the abbreviations used throughout the manuscript.

C. Proofs of the Scalarization Properties

Proof of Proposition 2.

PROOF. Let S^* be an optimal solution of $\min_{S\in\mathcal{S}} Z_\lambda(S)$, and suppose that it is not Pareto efficient. Then there exists $S' \in \mathcal{S}$ with $C_F(S') \le C_F(S^*)$ and $P(S') \le P(S^*)$, at least one inequality being strict. Since $\Delta_C > 0$ and $\Delta_P > 0$, the normalized criteria (39) satisfy $\overline{C}(S') \le \overline{C}(S^*)$ and $\overline{P}(S') \le \overline{P}(S^*)$, again with at least one strict inequality. Because $0 < \lambda < 1$, both weights $1-\lambda$ and λ are positive, so the maximum term in (40) does not increase, and the augmentation term strictly decreases because $\rho > 0$. Hence $Z_\lambda(S') < Z_\lambda(S^*)$, which contradicts the optimality of S^*.

Invariance under a common scaling of priorities. Let $p'_j = \gamma p_j$ for all $j \in J$ and some $\gamma > 0$. Then $P'(S) = \gamma P(S)$ for every $S \in \mathcal{S}$, while $C_F(S)$ is unchanged. The lexicographic anchor problems (37)–(38) therefore have the same optimal solutions, so that $\widehat{P}'_{\min} = \gamma \widehat{P}_{\min}$, $\widehat{P}'_C = \gamma \widehat{P}_C$,

Table A.1
Notation used in the priority-weighted LLRP.

Sets and indices	
I	Set of candidate depot locations, indexed by i
J	Set of pharmacies, indexed by j
$V = I \cup J$	Set of all nodes, indexed by u, v
A	Set of directed arcs, excluding depot-to-depot arcs: $A = \{(u,v) : u,v \in V,\ u \neq v\} \setminus (I \times I)$
A_J	Set of directed pharmacy-to-pharmacy arcs: $A_J = \{(u,v) : u,v \in J,\ u \neq v\}$
A^i	Set of arcs that a route of depot i may use: $A^i = \{(i,j),(j,i) : j \in J\} \cup A_J$
S	Set of feasible solutions of model (3)–(25), indexed by S
Parameters	
f_i	Fixed cost of opening candidate depot $i \in I$
F_i	Fixed deployment cost of a vehicle operating from depot $i \in I$
d_j	Deterministic demand of pharmacy $j \in J$
W_i	Capacity of depot $i \in I$, in demand units
Q	Capacity of each vehicle
N_g	Required number of open depots
N_v	Maximum number of vehicles available in the network
L	Maximum route duration, including the return trip to the depot
$c_{uv},\ \ell_{uv}$	Travel cost (road distance) and travel duration of arc $(u,v) \in A$
$\underline{\ell}_{uv}$	Duration of a shortest directed path from node u to node v
$e_j,\ L_j$	Earliest arrival time $e_j = \min_{i \in I} \underline{\ell}_{ij}$ and latest arrival time $L_j = L - \min_{i \in I} \underline{\ell}_{ji}$ at pharmacy j
M^T_{uv}	Big-M coefficient of the arrival-time constraints; $L + \ell_{uv}$, reduced to $\max\{0,\ \ell_{uv} + L_u - e_v\}$ in Section 3.4
p_j	Positive priority coefficient of pharmacy $j \in J$
Π	Total priority weight, $\Pi = \sum_{j \in J} p_j$
λ	Trade-off parameter of the Tchebycheff scalarization, $0 < \lambda < 1$ ($\lambda = 0,\ \lambda = 1$: cost and latency anchors)
ρ	Positive augmentation coefficient of the Tchebycheff scalarization
$\widehat{C}_{\min},\ \widehat{P}_C$	Cost anchor: minimum financial cost, and minimum latency among the solutions with this cost
$\widehat{P}_{\min},\ \widehat{C}_P$	Latency anchor: minimum latency, and minimum financial cost among the solutions with this latency
$\Delta_C,\ \Delta_P$	Normalization ranges, $\Delta_C = \widehat{C}_P - \widehat{C}_{\min}$ and $\Delta_P = \widehat{P}_C - \widehat{P}_{\min}$
Decision variables	
y_i	1 if candidate depot i is opened; 0 otherwise
z_{ij}	1 if pharmacy j is assigned to depot i; 0 otherwise
x^i_{uv}	1 if arc $(u,v) \in A^i$ is traversed by a route of depot i; 0 otherwise
X_{uv}	Aggregated flow on arc $(u,v) \in A_J$, $X_{uv} = \sum_{i \in I} x^i_{uv}$
k_i	Number of vehicles dispatched from depot i
ω_j	Cumulative vehicle load upon arrival at pharmacy j
t_j	Arrival-time variable of pharmacy j, with $t_j \geq a_j(S)$
$g_{ij},\ g_{uv}$	Load carried on depot arc (i,j) and on arc $(u,v) \in A_J$ in the commodity-flow formulation
$r_{ij},\ r_{uv}$	Priority flow on depot arc (i,j) and on arc $(u,v) \in A_J$ (total priority of the pharmacies still to be served)
ζ	Auxiliary variable equal to the larger weighted normalized deviation in the Tchebycheff formulation
Solution measures and objectives	
$a_j(S)$	Arrival time of pharmacy j in solution S (cumulative travel duration from its depot)
$C_F(S)$	Financial cost of solution S: depot-opening, vehicle-deployment, and travel costs
$P(S)$	Priority-weighted latency of solution S, $P(S) = \sum_{j \in J} p_j a_j(S)$
P^F	Priority-flow latency, equal to $P(S)$ for every integer solution (Proposition 1)
$\overline{C}(S),\ \overline{P}(S)$	Normalized financial cost and priority-weighted latency
$Z_\lambda(S)$	Augmented weighted Tchebycheff objective
Branch-and-price (Appendix D)	
D	Set of open depots, $D \subseteq I$, $\lvert D \rvert = N_g$
$\mathcal{R}_i,\ \mathcal{R}$	Set of feasible routes based at depot $i \in D$, and $\mathcal{R} = \bigcup_{i \in D} \mathcal{R}_i$
a_{jr}	1 if route r visits pharmacy j; 0 otherwise
d_r	Total demand served by route r, $d_r = \sum_{j \in J} d_j a_{jr}$
$c^F_r,\ p^L_r$	Financial cost (travel cost and vehicle cost) and priority-weighted latency of route r
θ_r	1 if route r is selected; 0 otherwise
Heuristic (Section 4)	
$F_o(S)$	Objective of solution S in optimization mode o: $C_F(S)$, $P(S)$, or $Z_\lambda(S)$
$\tau(r)$	Duration of route r, including the return trip to its depot
κ_{RCL}	Size of the restricted candidate list in ECWA
β	Penalty coefficient of vehicle-capacity violations in strategic oscillation

Table B.1
Abbreviations used throughout the manuscript.

Abbreviation	Meaning
BKS	Best-known solution
CPU	Central processing unit (computing time)
DR	Depot replacement
ECWA	Extended Clarke–Wright algorithm
GRASP	Greedy randomized adaptive search procedure
LB / UB	Lower bound / upper bound
LLRP	Latency location-routing problem
LP	Linear programming
LRP	Location-routing problem
MILP	Mixed-integer linear programming
MTZ	Miller–Tucker–Zemlin
RCL	Restricted candidate list
SO	Strategic oscillation
VND	Variable neighborhood descent

and the cost anchors are unchanged. Substituting into (39) gives $\overline{P}'(S) = \overline{P}(S)$ and $\overline{C}'(S) = \overline{C}(S)$, hence $Z'_\lambda(S) = Z_\lambda(S)$ for every $S \in \mathcal{S}$.

D. Branch-and-Price for the Anchor Problems

The branch-and-price algorithm solves the anchor problems of the larger networks for each set $D \subseteq I$ of N_g open depots; the optimal anchor value is the smallest value over all depot sets. For a fixed D, let $\mathcal{R}_i$ be the set of routes based at depot $i \in D$ that respect the vehicle capacity and the duration limit, let $\mathcal{R} = \bigcup_{i\in D} \mathcal{R}_i$, let $a_{jr} = 1$ if route r visits pharmacy j, and let $d_r = \sum_{j\in J} d_j a_{jr}$. The financial cost and the priority-weighted latency of route $r = (i, j_1, \dots, j_h, i)$ are

$$c_r^F = c_{ij_1} + \sum_{s=1}^{h-1} c_{j_s j_{s+1}} + c_{j_h i} + F_i, \qquad p_r^L = \sum_{s=1}^{h} p_{j_s} a_{j_s}(r), \tag{D.1}$$

where $a_{j_s}(r)$ is the arrival time at j_s along the route. With the binary variable θ_r selecting route r and the route value $b_r \in \{c_r^F, p_r^L\}$ of the criterion being minimized, the master problem is

$$\min \quad \sum_{r\in\mathcal{R}} b_r \theta_r \tag{D.2}$$

$$\text{s.t.} \quad \sum_{r\in\mathcal{R}} a_{jr}\theta_r = 1 \qquad \forall j \in J, \tag{D.3}$$

$$\sum_{r\in\mathcal{R}_i} d_r \theta_r \le W_i \qquad \forall i \in D, \tag{D.4}$$

$$1 \le \sum_{r\in\mathcal{R}_i} \theta_r \le N_v \qquad \forall i \in D, \tag{D.5}$$

$$\sum_{r\in\mathcal{R}} \theta_r \le N_v, \tag{D.6}$$

$$\theta_r \in \{0,1\} \qquad \forall r \in \mathcal{R}. \tag{D.7}$$

The opening costs $\sum_{i\in D} f_i$ are constant for a fixed depot set. In the second stage of a lexicographic anchor, the other criterion is minimized and the value of the first stage is imposed by a budget constraint, $\sum_r c_r^F \theta_r \le \widehat{C}_{\min} - \sum_{i\in D} f_i + 10^{-6}$ for the cost anchor and $\sum_r p_r^L \theta_r \le \widehat{P}_{\min} + 10^{-6}$ for the latency anchor.

The linear relaxation of (D.2)–(D.7) is solved by column generation. Let π_j, $\sigma_i \le 0$, μ_i and $\phi \le 0$ be the dual values of the covering, depot-capacity, route-count, and fleet constraints, where μ_i combines the duals of the two route-count inequalities. The reduced cost of route $r \in \mathcal{R}_i$ is

$$\bar{b}_r = b_r - \sum_{j\in J} a_{jr}\pi_j - d_r\sigma_i - \mu_i - \phi, \tag{D.8}$$

complemented, in the second stage, by the dual value of the budget constraint multiplied by the route value of the budgeted criterion. For each open depot, the pricing problem is an elementary shortest-path problem with capacity and duration resources. When the latency is part of the reduced cost, it is solved by a forward labeling algorithm in which a label stores the reduced cost, the load, the elapsed time, and the visited pharmacies, and the term $p_j a_j(r)$ is added when pharmacy j is reached. Since all priorities are nonnegative, a label is dominated by a label of the same node with smaller or equal reduced cost, load and elapsed time, and with a subset of the visited pharmacies. Elementarity is relaxed by ng-routes with neighborhoods of the eight nearest pharmacies (Baldacci et al., 2011), and subset-row inequalities on triplets of pharmacies are separated once column generation has converged (Jepsen et al., 2008). When only the financial cost is minimized, the elapsed time does not enter the reduced cost, and an elementary bidirectional labelling algorithm is used instead (Righini and Salani, 2006). The pricing problems of the open depots are solved in parallel.

Because a solution uses at most N_v routes, $z_{\text{RMP}} + N_v \min\{0, \min_r \bar{b}_r\}$, where z_{RMP} is the value of the restricted master problem, is a valid lower bound at every column-generation iteration (Lübbecke and Desrosiers, 2005); it allows a node to be pruned before column generation has converged. If the linear solution is fractional, the algorithm branches on the arc whose aggregated flow is closest to 0.5. In one branch the arc is forbidden; in the other it is imposed, and all other arcs leaving its tail and entering its head are forbidden, restricted to the pharmacy end for arcs incident to a depot, so that a depot may still operate several routes. Nodes are explored in best-first order. Feasible solutions are obtained by solving the restricted master problem with binary variables over the generated columns, and every incumbent is verified from its routes.

The depot sets are solved in parallel and share their incumbent, so that a depot set is discarded as soon as its lower bound reaches the best value found for any depot set. The smallest bound over the remaining depot sets is a valid lower bound on the anchor value. The algorithm is implemented in Python with Gurobi as the linear programming solver and with pricing routines compiled by Numba. Each depot set is given a time limit of two hours, which is checked between

column-generation rounds and can therefore be exceeded by a difficult depot set; the reported computing time of an anchor problem is that of the slowest depot set.

E. ECWA Construction Details

ECWA starts from a depot set D with $|D| = N_g$. For each pharmacy, the feasible depots are those in D whose capacity is not smaller than the demand of the pharmacy and from which the round trip to the pharmacy satisfies the duration limit L; the round trip is computed with the directed durations in both directions. The pharmacies are assigned in increasing order of the number of feasible depots and, among pharmacies with the same number, in decreasing order of demand. Each pharmacy is assigned to the feasible depot with sufficient remaining capacity that minimizes the round-trip duration multiplied by $1 + 0.35\,u^2$, where u is the utilization of the depot after the assignment; this factor prevents central depots from being filled before the pharmacies with few alternatives are assigned. Each assigned pharmacy forms a route of its own. If a pharmacy cannot be assigned, the construction fails and the iteration is discarded.

In the merging phase, every ordered pair of routes whose combined demand does not exceed Q is examined. The merged route is formed in the four combinations of forward and reversed sequences of the two routes, and it may be assigned to the depot of either route, provided that the depot capacity is respected. Each candidate is checked against the duration limit, and its saving is $s(r_1, r_2) = \tau(r_1) + \tau(r_2) - \tau(r_m)$. For each pair, the best feasible orientation and depot are retained. A pair is admissible if its saving is positive or if the number of routes still exceeds N_v, so that merges with nonpositive savings are used only to reach the fleet limit. One merge is drawn uniformly from the κ_{RCL} admissible merges with the largest savings, and the procedure is repeated until no admissible merge remains. If more than N_v routes remain at the end, the solution is infeasible and can become an incumbent only if VND-SO restores its feasibility. The construction does not use the priority coefficients, which enter only through the objective of the improvement phase.

F. Neighborhood Operators

The seven neighborhoods of VND-SO are explored in the following order.

1. Relocate removes a pharmacy from its position and reinserts it at another position of the same route or of another route.
2. Swap exchanges two pharmacies of the same route or of different routes; it also exchanges the depots of two routes based at different depots.
3. 2-opt reverses a segment of a route, and 2-opt* exchanges the final segments of two routes.
4. 2-relocate removes two consecutive pharmacies and reinserts them, in the same order, at another position of the same route or of another route.
5. Node–arc swap exchanges a pharmacy with a pair of consecutive pharmacies, within a route or between two routes.
6. Arc–arc swap exchanges two pairs of consecutive pharmacies, within a route or between two routes.
7. Swap* removes one pharmacy from each of two routes and inserts each of them into the other route at its best position.

G. Strategic Oscillation and Incumbent Management

During VND-SO, moves are evaluated with the penalized objective (47) of Section 4.3, in which only the vehicle capacity is relaxed; the depot capacities and the route-duration limit are always respected. The penalty coefficient is initialized from the objective value of the constructed solution divided by the total demand $D_{\mathrm{tot}} = \sum_{j \in J} d_j$, so that its scale follows that of the objective. In the financial and latency modes, it is bounded from below by one; in the Tchebycheff mode, whose normalized objective is much smaller, the absolute value of the objective is used and the lower bound is 10^{-6}, so that the coefficient remains positive even if a normalized value is negative because an anchor is not exact.

A move from S to S', including a depot replacement, is accepted if $\Phi_{o,\beta}(S') < \Phi_{o,\beta}(S) - \varepsilon_{\mathrm{move}}$, with $\varepsilon_{\mathrm{move}} = 10^{-8}$ in the Tchebycheff mode and 10^{-4} in the two anchor modes. Depot replacement evaluates every exchange of an open depot with a closed one, in which all routes of the open depot are moved to the closed depot, and applies the best accepted exchange; since all candidate depots of the case studies have the same opening and vehicle costs, these costs do not change. The route-level neighborhoods apply the first accepted move.

After each accepted move, the current solution is checked for feasibility. Two counters record the numbers of consecutive accepted moves that end in a feasible and in an infeasible solution. When four consecutive moves end in infeasible solutions, β is replaced by $\min\{\beta_{\max}, \xi\beta\}$; when four consecutive moves end in feasible solutions, it is replaced by $\max\{\beta_{\min}, \beta/\xi\}$. In both cases, ξ is drawn at random from $\{1.5, 2.5\}$ and the corresponding counter is reset; the bounds are $\beta_{\min} = 10^{-6}$ and $\beta_{\max} = 10^{9}$. Increasing β drives the search back to the feasible region, and decreasing it allows the search to cross infeasible solutions again.

The best feasible solution S_{best} of VND-SO is maintained separately from the current solution and is evaluated with the unpenalized objective. A feasible solution S replaces it if $F_o(S) < F_o(S_{\mathrm{best}}) - 10^{-4}$, or if no feasible solution has been found before. VND-SO returns S_{best} if it exists, and otherwise the current solution, which the GRASP framework rejects if it is infeasible. In Algorithm 1, the solutions returned by VND-SO are compared without tolerance.

Table I.1
Best latency over 30 runs by instance and search configuration.

Instance	Base	Base+SO	Base+SO+DR	GRASP-VND-SO
Ch50-5	1,661.64	1,663.45	1,661.64	1,661.64
Ch75-10	2,485.49	2,466.53	2,411.93	2,387.81
Ch100-10	3,833.17	3,808.72	3,839.32	3,822.38
Gaskell21-5	653.48	653.48	653.48	653.48
Gaskell29-5	1,199.33	1,199.33	1,199.33	1,199.33
Gaskell32-5	1,552.84	1,552.84	1,552.84	1,552.84
Gaskell36-5	1,627.17	1,627.17	1,627.17	1,627.17
Min27-5	5,387.55	5,387.55	5,387.55	5,387.55
Min134-8	22,649.74	23,082.11	22,621.33	22,324.46
Or117-14	54,219.84	53,824.54	54,047.47	53,824.54

H. VND-SO Pseudocode

Algorithm 2 VND-SO

Input: solution S, mode o, neighborhoods $\mathcal{N}_1, \dots, \mathcal{N}_7$
Output: best feasible solution S_{best}, or S if none is found

$\beta \leftarrow \beta_0(o, S)$; $c_{\text{feas}} \leftarrow 0$; $c_{\text{inf}} \leftarrow 0$; $S_{\text{best}} \leftarrow \emptyset$
if S is feasible **then**
    $S_{\text{best}} \leftarrow S$
end if
repeat
    $improved \leftarrow$ false
    if DEPOTREPLACE(S, o, β) finds an accepted move **then**
        $improved \leftarrow$ true; UPDATESTATE
    end if
    for $k = 1, \dots, 7$ **do**
        while FIRSTIMPROVE$(S, \mathcal{N}_k, o, \beta)$ finds an accepted move **do**
            $improved \leftarrow$ true; UPDATESTATE
        end while
    end for
until $improved =$ false
if $S_{\text{best}} \neq \emptyset$ **then**
    return S_{best}
else
    return S
end if

UPDATESTATE updates S_{best}, the counters c_{feas} and c_{inf} and the penalty coefficient β as described in Appendix G.

I. Benchmark Ablation: Instance-Level Results

Table I.1 reports the best latency over 30 runs for each instance and each of the four configurations of Section 5.2.2.

J. Anchor Values

Table J.1 reports the anchors used for the normalization of the 20 city–profile settings, together with the ranges $\Delta_C = \widehat{C}_P - \widehat{C}_{\min}$ and $\Delta_P = \widehat{P}_C - \widehat{P}_{\min}$. Each anchor is the lexicographically best solution of 30 heuristic runs in the corresponding mode; the cost runs are shared by the five profiles of a network, and their latencies are recomputed for each profile. The anchors of Afyon and Canakkale coincide with the exact values of Table 6, whereas those of Aydin deviate from them as discussed in Section 5.4.1.

K. Preference Paths

Table K.1 reports the travel cost C_T and the priority-weighted latency P of the anchors and of the best solutions for $\lambda \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$. Since all solutions of a network open N_g depots and use N_v vehicles, the financial cost is $C_F = C_T + 5{,}000 N_g + 200 N_v$.

Table J.1
Anchors used for the normalization of the 20 city–profile settings.

City	Priority profile	$\widehat{C}_{\min}$	$\widehat{P}_C$	$\widehat{C}_P$	$\widehat{P}_{\min}$	Δ_C	Δ_P
Afyon	Equal-Priority	10,859.963	18,594.00	10,896.869	7,791.00	36.906	10,803.00
Afyon	Tiered	10,859.963	11,864.70	10,874.510	4,298.70	14.547	7,566.00
Afyon	Demand-Correlated	10,859.963	10,211.67	10,877.462	4,317.85	17.499	5,893.82
Afyon	Negative-Correlated	10,859.963	17,679.33	10,868.306	6,856.28	8.343	10,823.05
Afyon	Hyper-Critical	10,859.963	26,964.00	10,873.529	12,641.00	13.566	14,323.00
Aydin	Equal-Priority	16,757.591	140,352.00	16,849.745	96,453.00	92.154	43,899.00
Aydin	Tiered	16,757.591	79,756.80	16,871.264	53,048.80	113.673	26,708.00
Aydin	Demand-Correlated	16,757.591	88,213.96	16,849.745	60,458.37	92.154	27,755.59
Aydin	Negative-Correlated	16,757.591	122,314.04	16,849.745	84,221.13	92.154	38,092.91
Aydin	Hyper-Critical	16,757.591	250,944.00	16,999.509	147,558.00	241.918	103,386.00
Canakkale	Equal-Priority	11,031.218	20,505.00	11,062.137	16,511.00	30.919	3,994.00
Canakkale	Tiered	11,031.218	12,513.30	11,041.097	10,158.70	9.879	2,354.60
Canakkale	Demand-Correlated	11,031.218	13,265.67	11,064.790	11,083.58	33.572	2,182.10
Canakkale	Negative-Correlated	11,031.218	17,491.83	11,055.721	13,635.25	24.503	3,856.57
Canakkale	Hyper-Critical	11,031.218	28,236.00	11,062.996	17,251.00	31.778	10,985.00
Isparta	Equal-Priority	11,263.775	49,168.00	11,295.664	29,294.00	31.889	19,874.00
Isparta	Tiered	11,263.775	32,172.70	11,295.664	17,897.30	31.889	14,275.40
Isparta	Demand-Correlated	11,263.775	26,742.63	11,287.257	16,340.61	23.482	10,402.02
Isparta	Negative-Correlated	11,263.775	47,009.37	11,289.449	27,552.02	25.674	19,457.36
Isparta	Hyper-Critical	11,263.775	67,969.00	11,293.156	36,623.00	29.381	31,346.00

Notes: $\widehat{C}_{\min}$ and $\widehat{C}_P$: financial cost; $\widehat{P}_C$ and $\widehat{P}_{\min}$: priority-weighted latency (s). Latencies of different profiles refer to different priority vectors and are not comparable.

Table K.1
Travel cost and priority-weighted latency along the preference paths.

	Equal		Tiered		Demand-correlated		Negative-correlated		Hyper-critical	
	C_T	P	C_T	P	C_T	P	C_T	P	C_T	P
Afyon										
Cost anchor	59.963	18,594	59.963	11,864.7	59.963	10,211.7	59.963	17,679.3	59.963	26,964
$\lambda = 0.1$	61.519	12,223	61.036	9,880.8	61.267	7,908.3	60.676	15,395.6	60.676	23,273
$\lambda = 0.3$	62.242	9,101	61.990	6,457.3	61.990	5,991.7	61.519	10,945.7	61.847	17,100
$\lambda = 0.5$	63.353	8,769	62.603	5,825.8	63.353	5,268.6	61.990	9,444.8	62.242	14,780
$\lambda = 0.7$	64.336	8,159	64.336	4,892.3	64.336	4,958.7	62.242	8,179.2	63.353	13,701
$\lambda = 0.9$	66.578	7,994	67.085	4,747.2	69.685	4,562.1	64.336	7,279.8	65.365	13,356
Latency anchor	96.869	7,791	74.510	4,298.7	77.462	4,317.9	68.306	6,856.3	73.529	12,641
Aydin										
Cost anchor	757.591	140,352	757.591	79,756.8	757.591	88,214.0	757.591	122,314.0	757.591	250,944
$\lambda = 0.1$	754.784	104,004	762.258	64,519.5	754.588	64,033.0	754.784	91,966.5	762.062	164,669
$\lambda = 0.3$	754.784	104,004	754.588	57,851.6	765.437	67,813.2	754.784	91,966.5	765.157	163,804
$\lambda = 0.5$	754.784	104,004	772.640	57,798.4	756.280	64,018.1	754.784	91,966.5	792.654	163,779
$\lambda = 0.7$	791.225	102,179	792.209	56,385.7	806.170	66,481.6	786.460	92,314.3	805.476	155,297
$\lambda = 0.9$	810.068	99,250	815.061	53,573	803.727	62,731.4	804.864	87,823.4	790.958	147,486
Latency anchor	849.745	96,453	871.264	53,048.8	849.745	60,458.4	849.745	84,221.1	999.509	147,558
Canakkale										
Cost anchor	31.218	20,505	31.218	12,513.3	31.218	13,265.7	31.218	17,491.8	31.218	28,236
$\lambda = 0.1$	32.523	18,182	31.983	11,789.3	32.707	11,903.0	32.498	15,413.1	33.736	25,403
$\lambda = 0.3$	34.272	17,495	32.891	11,127.1	34.509	11,636.2	34.247	14,654.2	35.780	20,948
$\lambda = 0.5$	36.806	17,235	34.399	10,900.9	36.105	11,374.5	35.950	14,392.0	36.951	19,188
$\lambda = 0.7$	39.349	16,969	35.812	10,625.6	36.919	11,224.1	37.895	14,156.4	39.915	18,671
$\lambda = 0.9$	54.045	16,826	37.124	10,301.2	38.449	11,129.1	51.441	13,988.1	48.946	17,883
Latency anchor	62.137	16,511	41.097	10,158.7	64.790	11,083.6	55.721	13,635.2	62.996	17,251
Isparta										
Cost anchor	63.775	49,168	63.775	32,172.7	63.775	26,742.6	63.775	47,009.4	63.775	67,969
$\lambda = 0.1$	66.190	42,928	66.073	27,099.4	65.452	23,040.7	65.253	38,477.4	65.899	57,002
$\lambda = 0.3$	69.003	36,446	68.449	22,821.3	67.885	20,595.0	67.879	34,528.7	68.584	48,867
$\lambda = 0.5$	70.667	33,579	70.618	20,759.6	69.857	19,104.5	69.806	32,008.7	71.378	44,732
$\lambda = 0.7$	72.971	31,755	73.372	19,710.2	72.591	18,036.2	71.740	30,161.2	73.805	41,077
$\lambda = 0.9$	77.100	30,173	77.136	18,561.2	74.802	16,841.4	76.034	28,553.8	76.559	37,867
Latency anchor	95.664	29,294	95.664	17,897.3	87.257	16,340.6	89.449	27,552.0	93.156	36,623

Notes: C_T: travel cost (km); P: priority-weighted latency (s). Interior entries are the best of 30 runs.

L. Additional Robustness Results

Table L.1 reports the robustness statistics for the remaining 19 city–profile combinations at five representative preference values. Each setting is evaluated over 30 independent runs, using the same experimental configuration as in the main computational study. Together with Table 10, these results cover all four networks and five priority profiles.

Table L.1: Robustness of GRASP-VND-SO over 30 runs for the remaining city–profile combinations.

City	Priority profile	λ	FR (%)	Z^*_λ	$\overline{Z}_\lambda$	$s_{Z,\lambda}$	Hits	Time (s)
Afyon	Tiered	0.1	100	0.0738	0.0744	0.0013	24	37.3
		0.3	100	0.0976	0.1002	0.0044	11	39.3
		0.5	100	0.1009	0.1022	0.0037	24	41.3
		0.7	100	0.0902	0.0902	0.0000	30	42.7
		0.9	100	0.0534	0.0534	0.0000	30	45.4
Afyon	Demand-Correlated	0.1	100	0.0671	0.0744	0.0049	5	37.6
		0.3	100	0.0852	0.0888	0.0030	12	41.4
		0.5	100	0.0969	0.0970	0.0003	27	41.9
		0.7	100	0.0761	0.0761	0.0000	30	42.9
		0.9	100	0.0556	0.0556	0.0000	24	36.5
Afyon	Negative-Correlated	0.1	100	0.0789	0.0813	0.0065	21	38.5
		0.3	100	0.1306	0.1361	0.0124	24	37.0
		0.5	100	0.1215	0.1316	0.0048	5	39.4
		0.7	100	0.0856	0.0868	0.0069	29	40.3
		0.9	100	0.0524	0.0524	0.0000	30	41.4
Afyon	Hyper-Critical	0.1	100	0.0742	0.0834	0.0125	19	40.0
		0.3	100	0.0972	0.1031	0.0055	14	40.3
		0.5	100	0.0840	0.0840	0.0000	30	42.6
		0.7	100	0.0750	0.0750	0.0000	30	47.5
		0.9	100	0.0449	0.0464	0.0017	17	42.8
Aydin	Equal-Priority	0.1	100	0.0172	0.4296	0.3342	1	682.9
		0.3	100	0.0516	0.3308	0.1650	2	691.7
		0.5	100	0.0860	0.4439	0.2587	1	668.0
		0.7	100	0.1095	0.3000	0.1239	1	610.7
		0.9	100	0.0574	0.2665	0.1284	1	666.4
Aydin	Tiered	0.1	100	0.0430	0.3763	0.2899	1	605.4
		0.3	100	0.0540	0.3365	0.1653	3	606.9
		0.5	100	0.0889	0.3199	0.1173	2	670.3
		0.7	100	0.0914	0.3165	0.0995	1	628.5
		0.9	100	0.0506	0.2950	0.1750	1	702.1
Aydin	Demand-Correlated	0.1	100	0.0129	0.5255	0.4163	1	759.0
		0.3	100	0.0795	0.3413	0.1986	1	769.9
		0.5	100	0.0641	0.3790	0.2412	2	771.7
		0.7	100	0.1582	0.3570	0.1459	1	744.9
		0.9	100	0.0737	0.2296	0.1193	1	727.6
Aydin	Negative-Correlated	0.1	100	0.0203	0.4800	0.3571	2	664.2
		0.3	100	0.0610	0.3682	0.1881	2	581.1
		0.5	100	0.1017	0.3022	0.1364	2	649.2
		0.7	100	0.1487	0.3338	0.1056	1	778.2
		0.9	100	0.0851	0.2377	0.0881	1	674.0
Aydin	Hyper-Critical	0.1	100	0.0166	0.1751	0.1473	1	664.1
		0.3	100	0.0471	0.1830	0.1082	2	542.3
		0.5	100	0.0785	0.1752	0.0634	1	563.8
		0.7	100	0.0594	0.1519	0.0555	1	595.1
		0.9	100	0.0138	0.0755	0.0551	1	673.8
Canakkale	Equal-Priority	0.1	100	0.0418	0.0419	0.0004	29	1,849.7
		0.3	100	0.0739	0.0940	0.0120	4	1,854.3
		0.5	100	0.0907	0.1050	0.0176	2	1,600.6
		0.7	100	0.0803	0.1139	0.0367	11	1,702.8
		0.9	100	0.0738	0.0894	0.0250	4	1,692.5
Canakkale	Tiered	0.1	100	0.0697	0.0713	0.0008	2	2,085.5
		0.3	100	0.1234	0.1315	0.0096	4	1,855.6
		0.5	100	0.1610	0.1783	0.0094	4	1,754.2
		0.7	100	0.1395	0.1518	0.0052	3	1,793.1
		0.9	100	0.0598	0.0695	0.0128	1	1,783.7
Canakkale	Demand-Correlated	0.1	100	0.0399	0.0416	0.0039	21	1,789.5
		0.3	100	0.0760	0.0869	0.0029	1	1,929.9
		0.5	100	0.0728	0.0736	0.0038	24	1,888.8
		0.7	100	0.0510	0.0579	0.0042	5	1,892.4
		0.9	100	0.0215	0.0462	0.0229	11	1,548.2
Canakkale	Negative-Correlated	0.1	100	0.0470	0.0491	0.0024	7	2,139.9
		0.3	100	0.0865	0.0937	0.0089	1	1,824.0
		0.5	100	0.0981	0.1071	0.0133	6	1,834.5
		0.7	100	0.0946	0.1090	0.0194	2	1,787.8
		0.9	100	0.0825	0.1029	0.0286	9	1,499.1
Canakkale	Hyper-Critical	0.1	100	0.0742	0.0742	0.0000	30	1,636.6
		0.3	100	0.1010	0.1071	0.0036	1	1,448.8

Continued on next page

Table L.1: Robustness of GRASP-VND-SO (continued).

City	Priority profile	λ	FR (%)	Z^*_λ	$\overline{Z}_\lambda$	$s_{Z,\lambda}$	Hits	Time (s)
		0.5	100	0.0902	0.0951	0.0039	3	1,429.5
		0.7	100	0.0905	0.0919	0.0009	4	1,814.7
		0.9	100	0.0558	0.0635	0.0108	9	1,327.2
Isparta	Equal-Priority	0.1	100	0.0686	0.0829	0.0065	1	5,733.4
		0.3	100	0.1148	0.1286	0.0104	1	5,015.6
		0.5	100	0.1081	0.1263	0.0109	1	4,909.3
		0.7	100	0.0867	0.1075	0.0113	1	4,935.4
		0.9	100	0.0418	0.0557	0.0079	1	4,679.6
Isparta	Tiered	0.1	100	0.0649	0.0801	0.0083	1	6,360.6
		0.3	100	0.1035	0.1278	0.0130	1	5,643.9
		0.5	100	0.1073	0.1262	0.0106	1	5,474.9
		0.7	100	0.0903	0.1083	0.0118	1	5,618.9
		0.9	100	0.0419	0.0543	0.0075	1	5,460.4
Isparta	Demand-Correlated	0.1	100	0.0644	0.0911	0.0127	1	5,796.2
		0.3	100	0.1227	0.1498	0.0107	1	5,228.3
		0.5	100	0.1329	0.1525	0.0131	1	4,973.2
		0.7	100	0.1141	0.1341	0.0091	1	5,208.8
		0.9	100	0.0470	0.0665	0.0126	1	5,132.7
Isparta	Negative-Correlated	0.1	100	0.0562	0.0864	0.0127	1	6,026.3
		0.3	100	0.1119	0.1468	0.0142	1	5,145.4
		0.5	100	0.1175	0.1457	0.0160	1	5,059.8
		0.7	100	0.0939	0.1221	0.0154	1	5,129.0
		0.9	100	0.0478	0.0675	0.0111	1	4,615.3
Isparta	Hyper-Critical	0.1	100	0.0651	0.0824	0.0100	1	5,956.1
		0.3	100	0.1172	0.1310	0.0074	1	5,141.2
		0.5	100	0.1294	0.1419	0.0058	1	4,959.1
		0.7	100	0.1024	0.1124	0.0059	1	5,208.6
		0.9	100	0.0435	0.0544	0.0050	1	4,874.7

Notes: FR is the percentage of feasible runs. Z^*_λ is the best recorded scalar objective value among the 30 runs for each setting. $\overline{Z}_\lambda$ and $s_{Z,\lambda}$ are the mean and sample standard deviation over feasible runs. Hits denotes the number of runs attaining Z^*_λ within a tolerance of 10^{-9}. Time denotes the mean recorded runtime in seconds. Each run uses 50 GRASP iterations. Values are rounded to four decimal places; a reported standard deviation of 0.0000 may therefore represent a small nonzero variation.

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